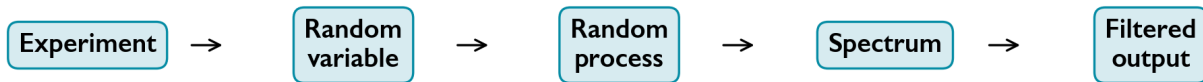


Random Signals and Noise



Probability, random signals, spectra and filtering.

Contents

- Random Variables and Probability Laws
- Moments and Transformations
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Symbols

Symbol	Meaning
X, x	Random variable and its value. $X(t)$ is a random process; $x(t)$ is a realization.
$f_X(x), F_X(x)$	Probability density and cumulative distribution.
μ_X, σ_X^2	Mean and variance.
$R_X(\tau), C_X(\tau)$	Autocorrelation and autocovariance.
$R_{XY}(\tau)$	$E\{X(t+\tau)Y(t)\}$: the first process carries the lag.
$S_X(f), S_{XY}(f)$	Power spectrum and cross spectrum; f is in hertz.

Symbol	Meaning
$h (t) , H (f)$	Impulse response and frequency response.
$N_0/2$	Two-sided white-noise spectral density.
B, W	Bandwidths defined locally. An ideal lowpass edge B corresponds to full positive-lobe bandpass width $2B$ where stated.

$$\text{sinc} (u) = \frac{\sin (\pi u)}{\pi u}, \qquad \text{sinc} (0) = 1.$$

Random Variables and Probability Laws

Events and random variables

A random experiment has an uncertain outcome. Its sample space S contains all possible outcomes; an event A is a subset of S .

$$P(S) = 1, \quad 0 \leq P(A) \leq 1,$$

$$P(A \cup B) = P(A) + P(B), \quad A \cap B = \emptyset.$$

If A occurs n_A times in n repeated trials, n_A/n is its observed relative frequency.

For a fair die, $S = \{1, 2, 3, 4, 5, 6\}$ and the even event is $A = \{2, 4, 6\}$, so $P(A) = 3/6 = 1/2$.

A random variable assigns a real value to each outcome:

$$X: S \rightarrow \mathbb{R}, \quad X(s_k) = x.$$

X denotes the variable; x denotes a possible value. A discrete variable has a finite or countable set of values. The continuous variables considered below possess probability densities.

Probability mass and cumulative distribution

The probability mass function (PMF) of a discrete variable is

$$p_X(x_k) = P(X = x_k), \quad p_X(x_k) \geq 0, \quad \sum_k p_X(x_k) = 1.$$

A Bernoulli variable has $P(X = 1) = p$ and $P(X = 0) = 1 - p$, where $0 \leq p \leq 1$.

The cumulative distribution function (CDF) is

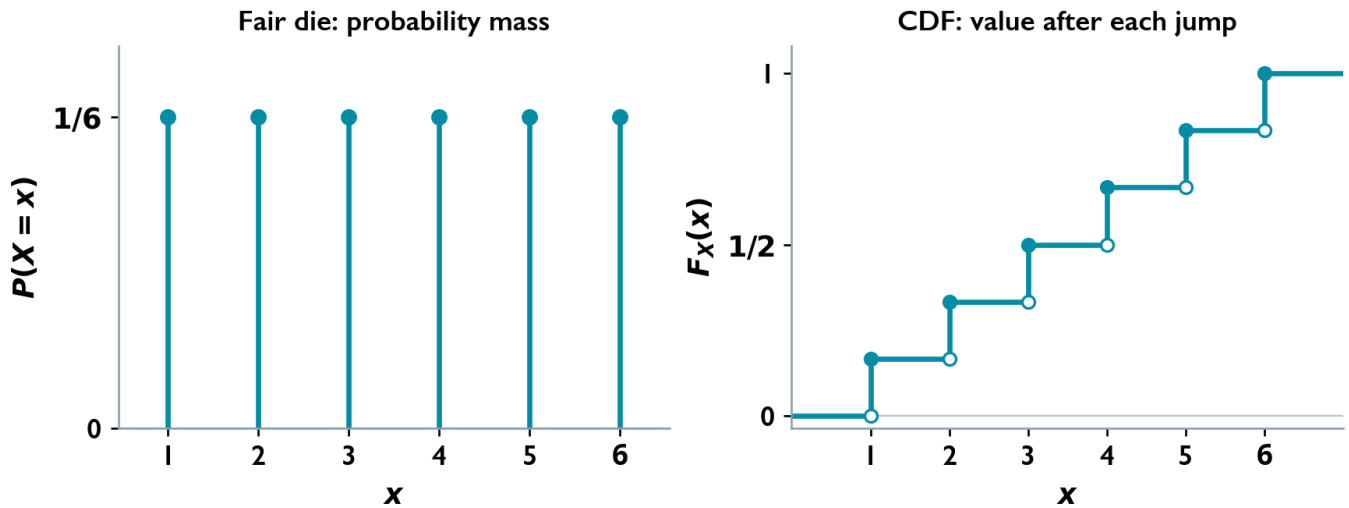
$$F_X(x) = P(X \leq x).$$

It is nondecreasing, lies in $[0, 1]$, and satisfies

$$F_X(-\infty) = 0, \quad F_X(+\infty) = 1, \quad P(a < X \leq b) = F_X(b) - F_X(a).$$

For a discrete variable, $F_X(x) = \sum_{x_k \leq x} p_X(x_k)$. A fair die has six masses of $1/6$, so its CDF increases by $1/6$ at each face value. A point mass is a CDF jump:

$P(X = x_0) = F_X(x_0) - F_X(x_0^-)$, where $F_X(x_0^-)$ is the left limit. A mixed law retains both its point masses and its continuous density.



Probability masses and CDF of a fair die.

Probability density

A probability density function (PDF) satisfies

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1,$$

$$F_X(x) = \int_{-\infty}^x f_X(u) du, \quad f_X(x) = \frac{dF_X(x)}{dx},$$

where the derivative exists. Thus

$$P(a < X \leq b) = \int_a^b f_X(x) dx.$$

A density is not a point probability: $P(X = x) = 0$. Its value may exceed one; its total area must equal one.

Uniform distribution

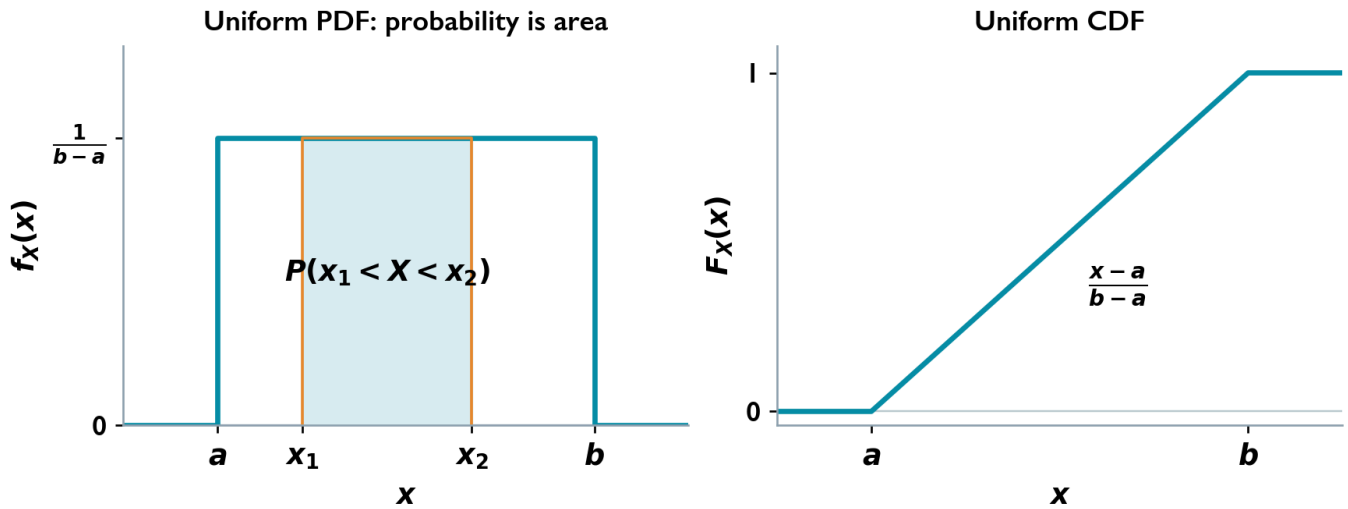
For X uniform on $[a, b]$, with $a < b$,

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

$$F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

For $a \leq c < d \leq b$,

$$P(c < X \leq d) = \frac{d - c}{b - a}.$$



Uniform PDF and CDF; the shaded area is an interval probability.

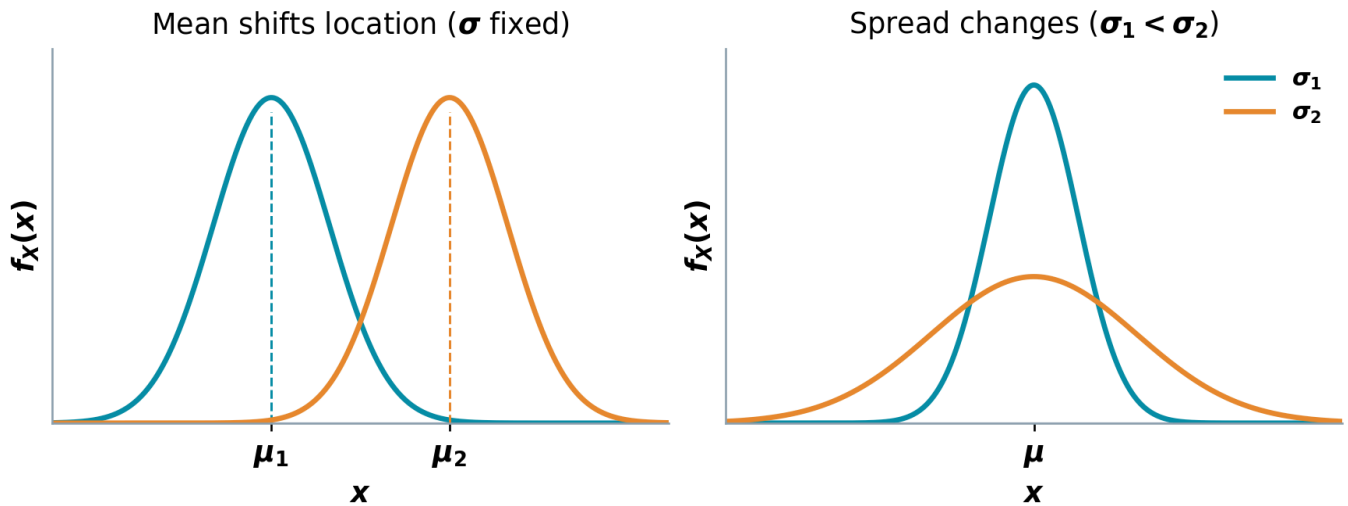
Gaussian distribution and Q function

A Gaussian variable $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ has mean μ_X , variance σ_X^2 , standard deviation $\sigma_X > 0$, and density

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma_X} \exp \left[-\frac{(x - \mu_X)^2}{2\sigma_X^2} \right], \quad x \in \mathbb{R}.$$

Standardization gives

$$Z = \frac{X - \mu_X}{\sigma_X} \sim \mathcal{N}(0, 1).$$



Standardization: $Z = (X - \mu)/\sigma$

Gaussian location, spread, and standardization.

Define the standard Gaussian upper tail by

$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^{\infty} e^{-u^2/2} du.$$

Then

$$P(X > c) = Q\left(\frac{c - \mu_X}{\sigma_X}\right), \quad Q(-z) = 1 - Q(z), \quad Q(0) = \frac{1}{2}.$$

Example: binary PAM decision error

Question

Binary symbols use levels $+A$ and $-A$, where $A > 0$. The received value is the transmitted level plus Gaussian noise $N \sim \mathcal{N}(0, \sigma_N^2)$, independent of the transmitted symbol. With decision threshold zero, find the error probability after transmitting $+A$.

Solution

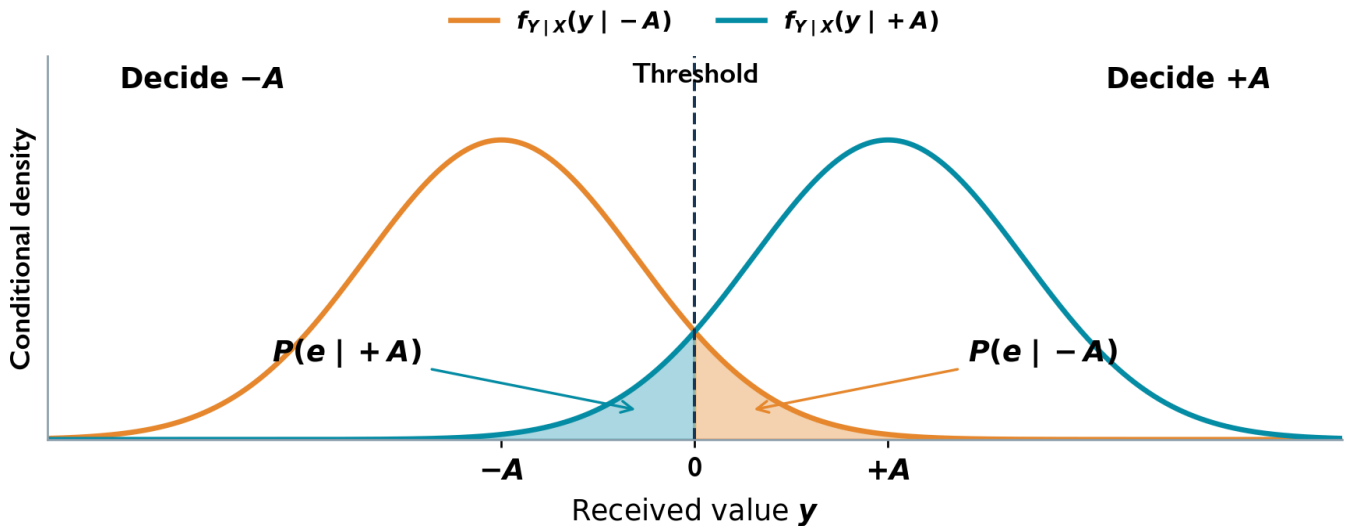
Given $+A$, $Y = A + N \sim \mathcal{N}(A, \sigma_N^2)$, and an error occurs when $Y < 0$:

$$P(\text{error} \mid +A) = \int_{-\infty}^0 \frac{e^{-(y-A)^2/(2\sigma_N^2)}}{\sqrt{2\pi} \sigma_N} dy.$$

With $z = (y - A)/\sigma_N$ and Gaussian symmetry,

$$P(\text{error} \mid +A) = P\left(Z < -\frac{A}{\sigma_N}\right) = Q\left(\frac{A}{\sigma_N}\right).$$

The conditional error after sending $-A$ is identical, so $P_e = Q(A/\sigma_N)$ for the symmetric binary model.



Zero-threshold PAM decision and Gaussian error tails.

Central limit theorem

Let $X_1, \dots, X_n$ be independent identically distributed variables with common finite mean μ_X and finite positive variance σ_X^2 . For their sum,

$$Y_n = \sum_{k=1}^n X_k, \quad E\{Y_n\} = n\mu_X, \quad \sigma_{Y_n}^2 = n\sigma_X^2.$$

As $n \rightarrow \infty$, the distribution of

$$Z_n = \frac{Y_n - n\mu_X}{\sigma_X \sqrt{n}}$$

approaches $\mathcal{N}(0, 1)$. This explains the Gaussian model for a sum of many small independent noise contributions. For finite n it is an approximation; independent Gaussian summands give an exactly Gaussian sum.

Worked Problems

A cubic cumulative distribution

Question

A random variable X has cumulative distribution

$$F_X(x) = \begin{cases} 0, & x < 0, \\ Ax^3, & 0 \leq x \leq 10, \\ B, & x > 10. \end{cases}$$

Determine A and B , and obtain and sketch the density.

Solution

Normalization and right continuity

$$\lim_{x \rightarrow \infty} F_X(x) = B = 1, \quad F_X(10) = 1000A = 1.$$

$$A = 10^{-3}, \quad B = 1.$$

Density The CDF is continuous, so differentiating gives

$$f_X(x) = \begin{cases} 3 \times 10^{-3}x^2, & 0 < x < 10, \\ 0, & \text{otherwise.} \end{cases}$$

Check and sketch

$$\int_0^{10} 3 \times 10^{-3}x^2 dx = 10^{-3} [x^3]_0^{10} = 1.$$

The density rises parabolically from zero to a limiting height of 0.3 on $(0, 10)$ and vanishes outside this interval.

Testing an exponential density

Question

Determine whether

$$g(x) = \begin{cases} e^{-x/2}, & x \geq 0, \\ 0, & x < 0 \end{cases}$$

is a valid probability density.

Solution

Nonnegativity $g(x) \geq 0$ for every x .

Total area

$$\int_{-\infty}^{\infty} g(x) dx = \int_0^{\infty} e^{-x/2} dx = \left[-2e^{-x/2} \right]_0^{\infty} = 2 \neq 1.$$

g is not a normalized probability density.

The corresponding normalized density is

$$f_X(x) = \frac{1}{2}e^{-x/2}u(x).$$

Normalizing a symmetric exponential law

Question

Find k such that

$$f_X(x) = ke^{-|x|}, \quad -\infty < x < \infty,$$

is a probability density.

Solution

Normalization Since $e^{-|x|} > 0$, require $k \geq 0$. **Split at the origin:**

$$1 = k \int_{-\infty}^0 e^x dx + k \int_0^{\infty} e^{-x} dx = k + k = 2k.$$

$$k = \frac{1}{2}, \quad f_X(x) = \frac{1}{2}e^{-|x|}.$$

By symmetry, each half-line has probability $1/2$.

A trapezoidal probability density

Question

Let

$$f_X(x) = \begin{cases} ax, & 0 \leq x \leq 1, \\ a, & 1 < x \leq 2, \\ a(3-x), & 2 < x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Determine a , obtain and sketch the CDF, and calculate $P(X \leq 1.5)$.

Solution

Normalization

$$1 = a \int_0^1 x \, dx + a \int_1^2 dx + a \int_2^3 (3 - x) \, dx = \frac{a}{2} + a + \frac{a}{2} = 2a.$$

$$a = \frac{1}{2}.$$

Accumulated probability

$$F_X(x) = \int_0^x \frac{t}{2} dt = \frac{x^2}{4}, \quad 0 \leq x < 1,$$

$$F_X(x) = \frac{1}{4} + \int_1^x \frac{1}{2} dt = \frac{x}{2} - \frac{1}{4}, \quad 1 \leq x < 2,$$

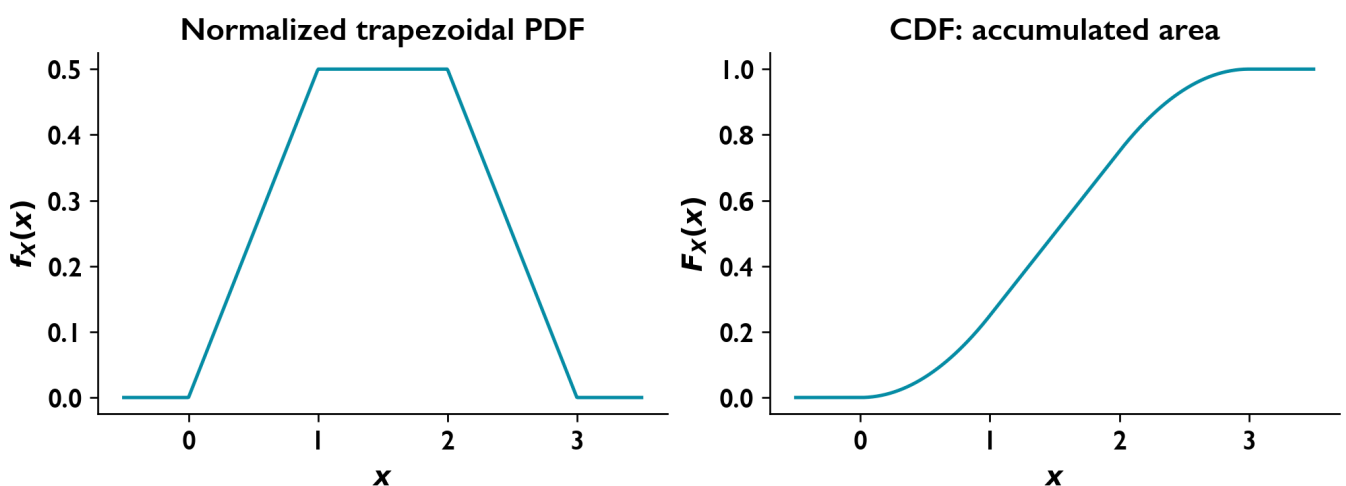
$$F_X(x) = 1 - \int_x^3 \frac{3-t}{2} dt = 1 - \frac{(3-x)^2}{4}, \quad 2 \leq x < 3.$$

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2/4, & 0 \leq x < 1, \\ x/2 - 1/4, & 1 \leq x < 2, \\ 1 - (3-x)^2/4, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$$

The CDF is continuous and increases quadratically, linearly, then along a concave quadratic segment.

Requested probability

$$P(X \leq 1.5) = F_X(1.5) = \frac{1.5}{2} - \frac{1}{4} = \frac{1}{2}.$$



The normalized trapezoidal density and its accumulated probability.

Normalizing a two-level density

Question

A density is specified by

$$f_X(x) = \begin{cases} k, & 0 < x < 1, \\ 1/3, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Determine k .

Solution

Total area

$$1 = \int_0^1 k \, dx + \int_1^2 \frac{1}{3} \, dx = k + \frac{1}{3}.$$
$$k = \frac{2}{3}.$$

Both density levels are nonnegative.

A shifted exponential law with a point mass

Question

A random voltage has generalized density

$$f_X(x) = k u(x+7) e^{-1.72(x+7)} + 0.35 \delta(x-1),$$

where u is the unit step and δ is the Dirac impulse.

- Determine the normalizing constant.
- Obtain and sketch the CDF.
- Calculate $P(X \leq 0.5)$, $P(X > 3)$, and $P(0.5 \leq X \leq 3)$.

Solution

Continuous and discrete probability The impulse assigns $P(X = 1) = 0.35$; the continuous part must integrate to 0.65.

$$1 = k \int_{-7}^{\infty} e^{-1.72(x+7)} \, dx + 0.35 = \frac{k}{1.72} + 0.35.$$

$$k = 1.72(0.65) = 1.118.$$

Cumulative distribution

$$F_X(x) = \begin{cases} 0, & x < -7, \\ 0.65 \left[1 - e^{-1.72(x+7)} \right], & -7 \leq x < 1, \\ 0.65 \left[1 - e^{-1.72(x+7)} \right] + 0.35, & x \geq 1. \end{cases}$$

The CDF rises from zero at -7 , jumps by 0.35 at 1 , and approaches one asymptotically.

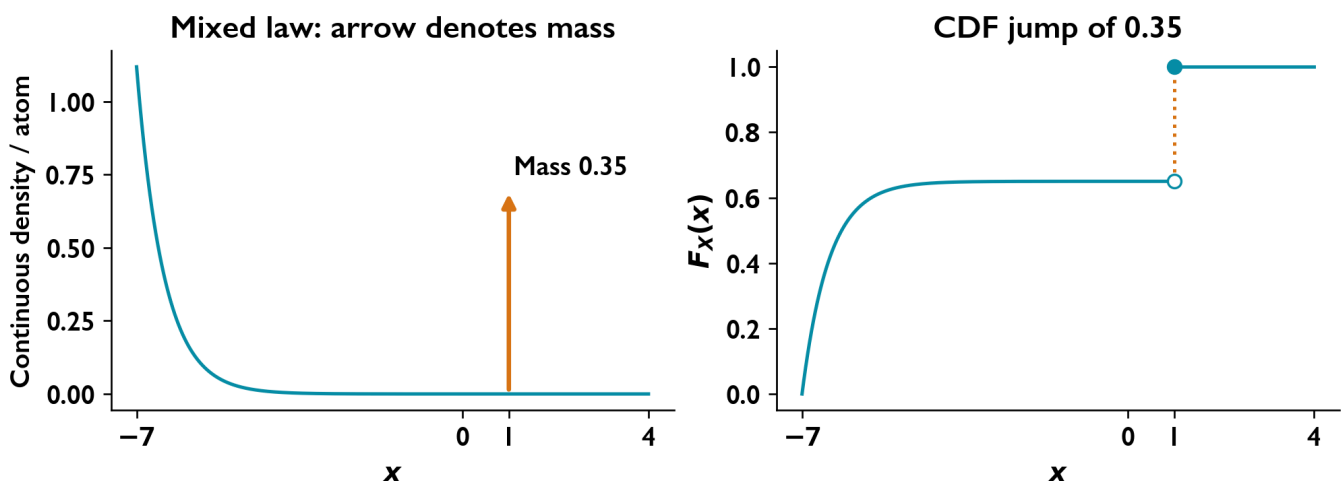
Requested probabilities There are no point masses at 0.5 or 3 .

$$P(X \leq 0.5) = 0.65 (1 - e^{-12.9}) \approx 0.6499983763.$$

$$P(X > 3) = 0.65e^{-17.2} \approx 2.20317 \times 10^{-8}.$$

$$\begin{aligned} P(0.5 \leq X \leq 3) &= F_X(3) - F_X(0.5) \\ &= 0.35 + 0.65 (e^{-12.9} - e^{-17.2}). \end{aligned}$$

$$P(0.5 \leq X \leq 3) \approx 0.3500016017.$$



The arrow represents a probability mass of 0.35 ; it is not an ordinary finite density height.

Cumulative probabilities of an Erlang density

Question

A continuous random variable has density

$$f_X(x) = xe^{-x}u(x).$$

Obtain its CDF and calculate $P(X \leq 1)$, $P(1 \leq X \leq 2)$, and $P(X > 2)$.

Solution

Cumulative distribution For $x \geq 0$, integration by parts gives

$$F_X(x) = \int_0^x t e^{-t} dt = \left[- (t+1) e^{-t} \right]_0^x = 1 - (1+x) e^{-x}.$$

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - (1+x) e^{-x}, & x \geq 0. \end{cases}$$

Its limit at infinity is one, verifying normalization.

Requested probabilities

$$P(X \leq 1) = 1 - \frac{2}{e} \approx 0.264241.$$

$$\begin{aligned} P(1 \leq X \leq 2) &= F_X(2) - F_X(1) \\ &= \frac{2}{e} - \frac{3}{e^2}. \end{aligned}$$

$$P(1 \leq X \leq 2) \approx 0.329753.$$

$$P(X > 2) = 1 - F_X(2) = \frac{3}{e^2} \approx 0.406006.$$

The cumulative distribution of a uniform variable

Question

For $a < b$, let

$$f_X(x) = \begin{cases} 1/(b-a), & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Obtain and sketch its CDF.

Solution

Integration over the support

$$F_X(x) = \int_a^x \frac{1}{b-a} dt = \frac{x-a}{b-a}, \quad a \leq x \leq b.$$

No probability lies below a , and all probability has accumulated by b .

$$F_X(x) = \begin{cases} 0, & x < a, \\ (x-a)/(b-a), & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

The sketch is horizontal at zero, rises linearly from $(a, 0)$ to $(b, 1)$ with slope $1/(b-a)$, then remains at one.

Recovering a density from a sinusoidal CDF

Question

A random variable has CDF

$$F_X(x) = \begin{cases} 0, & x < 0, \\ k \sin(\pi x/40), & 0 \leq x \leq 10, \\ k \sin(\pi/4), & x > 10. \end{cases}$$

Determine k , obtain the density, and calculate $P(X \leq 4)$.

Solution

Normalization

$$\lim_{x \rightarrow \infty} F_X(x) = k \sin(\pi/4) = 1.$$

$$k = \sqrt{2}.$$

Differentiation The CDF is continuous at both support boundaries, so no point masses occur there.

$$f_X(x) = \begin{cases} \frac{\sqrt{2}\pi}{40} \cos(\pi x/40), & 0 < x < 10, \\ 0, & \text{otherwise.} \end{cases}$$

The density is nonnegative on its support, and

$$\int_0^{10} f_X(x) dx = \sqrt{2} \sin(\pi/4) = 1.$$

Requested probability

$$P(X \leq 4) = \sqrt{2} \sin(\pi/10) \approx 0.437016.$$

Moments and Transformations

Mean, moments, and variance

The mean is a probability-weighted average:

$$\mu_X = E\{X\} = \sum_k x_k p_X(x_k) \quad \text{or} \quad \mu_X = \int_{-\infty}^{\infty} x f_X(x) dx.$$

For a function g ,

$$E\{g(X)\} = \sum_k g(x_k) p_X(x_k) \quad \text{or} \quad E\{g(X)\} = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

The raw moment of order n is $E\{X^n\}$. The first moment is the mean; the second is the mean-square value. For finite second moments,

$$\sigma_X^2 = E\{(X - \mu_X)^2\} = E\{X^2\} - \mu_X^2, \quad \sigma_X = \sqrt{\sigma_X^2}.$$

The RMS value is $\sqrt{E\{X^2\}}$. Mean square equals variance only for zero mean. For voltage across resistance R , average power is $E\{X^2\}/R$.

For constants c, d ,

$$E\{cX + d\} = c\mu_X + d, \quad \text{Var}(cX + d) = c^2\sigma_X^2.$$

For Bernoulli X , $X^2 = X$, giving $\mu_X = p$ and $\sigma_X^2 = p(1 - p)$.

Estimates from measurements

For N independent identically distributed observations $x_1, \dots, x_N$ with finite variance,

$$\hat{\mu}_X = \frac{1}{N} \sum_{n=1}^N x_n,$$

$$\hat{\sigma}_X^2 = \frac{1}{N-1} \sum_{n=1}^N (x_n - \hat{\mu}_X)^2, \quad N > 1.$$

These estimate the population mean and variance. The denominator $N - 1$ makes the variance estimate unbiased under the stated conditions.

Transforming a variable

For $Y = g(X)$, begin with

$$F_Y(y) = P(g(X) \leq y).$$

Suppose X has a density and g is strictly monotone on its support, with inverse $h = g^{-1}$. Then

$$F_Y(y) = \begin{cases} F_X(h(y)), & g \text{ increasing,} \\ 1 - F_X(h(y)), & g \text{ decreasing.} \end{cases}$$

Where the inverse is differentiable,

$$f_Y(y) = f_X(h(y)) \left| \frac{dh(y)}{dy} \right|.$$

Transform the support as well as the formula. If several inverse branches occur, each admissible branch contributes density. A constant transformation branch can create a point mass. With $A = \{x: g(x) = y_0\}$, its total mass is $P(Y = y_0) = P(X \in A) = F_Y(y_0) - F_Y(y_0^-)$, where the superscript minus denotes the left limit. Retain the remaining continuous density separately. A Gaussian input under $Y = cX + d$, $c \neq 0$, gives

$$Y \sim \mathcal{N}(c\mu_X + d, c^2\sigma_X^2).$$

Example: Gaussian voltage and averaging

Question

A zero-mean Gaussian voltage has standard deviation 10 V. Add 5 V. Find the probability of a positive reading and the probability that the average of two independent readings is positive.

Solution

Let $X \sim \mathcal{N}(0, 100)$ and $Y = X + 5$. Then

$$\mu_Y = 5, \quad \sigma_Y^2 = 100,$$

$$P(Y > 0) = Q\left(\frac{0-5}{10}\right) = 1 - Q(0.5) \approx 0.6915.$$

For $M = (Y_1 + Y_2)/2$, independence gives

$$\mu_M = 5, \quad \sigma_M^2 = \frac{100 + 100}{4} = 50,$$

$$P(M > 0) = Q\left(-\frac{5}{\sqrt{50}}\right) = 1 - Q\left(\frac{5}{\sqrt{50}}\right) \approx 0.7602.$$

Example: double-exponential density

Question

For $f_X(x) = ae^{-b|x|}$ on $\mathbb{R}$, determine the constants required for a density, find its CDF, and evaluate $P(1 < X < 2)$.

Solution

Integrability requires $b > 0$. Normalization gives

$$1 = 2a \int_0^\infty e^{-bx} dx = \frac{2a}{b}, \quad a = \frac{b}{2}.$$

Integrating separately on either side of zero gives

$$F_X(x) = \begin{cases} \frac{1}{2}e^{bx}, & x < 0, \\ 1 - \frac{1}{2}e^{-bx}, & x \geq 0. \end{cases}$$

Thus

$$P(1 < X < 2) = F_X(2) - F_X(1) = \frac{1}{2}(e^{-b} - e^{-2b}).$$

Example: Rayleigh distribution

Question

For $b > 0$, a Rayleigh variable has

$$f_R(r) = \begin{cases} \frac{r}{b}e^{-r^2/(2b)}, & r > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find its CDF, mean, mean square, and variance.

Solution

Integration gives

$$F_R(r) = \begin{cases} 0, & r < 0, \\ 1 - e^{-r^2/(2b)}, & r \geq 0. \end{cases}$$

Integration by parts reduces the mean to a Gaussian integral:

$$E\{R\} = \frac{1}{b} \int_0^\infty r^2 e^{-r^2/(2b)} dr = \int_0^\infty e^{-r^2/(2b)} dr = \sqrt{\frac{b\pi}{2}}.$$

With $u = r^2/(2b)$,

$$E\{R^2\} = \frac{1}{b} \int_0^\infty r^3 e^{-r^2/(2b)} dr = 2b \int_0^\infty u e^{-u} du = 2b.$$

Therefore

$$\sigma_R^2 = 2b - \frac{b\pi}{2} = b \left(2 - \frac{\pi}{2} \right).$$

Worked Problems

A transformation producing both a discrete mass and a continuous density

Question

Let $X \sim \mathcal{U}[-2, 2]$ and $Y = g(X)$, where

$$g(x) = \begin{cases} -1, & -2 \leq x < 0, \\ 2x/3, & 0 \leq x < 3, \\ 2, & x \geq 3. \end{cases}$$

Find and sketch $F_X(x)$, $f_Y(y)$, and $F_Y(y)$.

Solution

Input distribution

$$f_X(x) = \frac{1}{4} \mathbf{1}_{(-2, 2)}(x), \quad F_X(x) = \begin{cases} 0, & x < -2, \\ (x+2)/4, & -2 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

Point mass and continuous density The input never reaches $x = 3$, so no mass occurs at $y = 2$.

$$P(Y = -1) = P(X < 0) = \frac{1}{2},$$

$$f_{Y,\text{cont}}(y) = f_X(3y/2) \left| \frac{dx}{dy} \right| = \frac{13}{42} = \frac{3}{8}, \quad 0 < y < \frac{4}{3}.$$

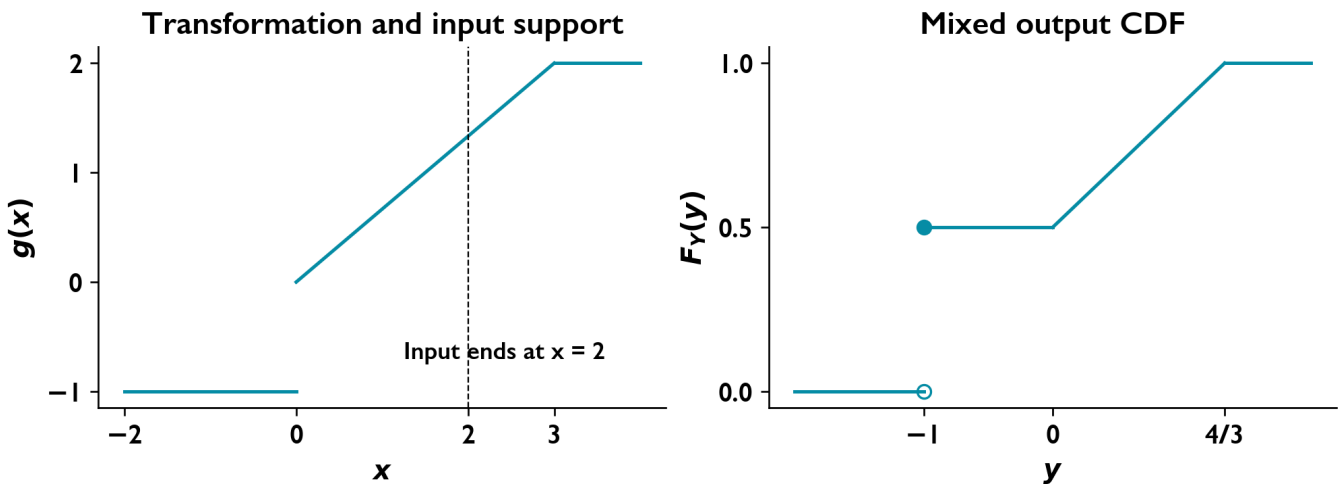
$$f_Y(y) = \frac{1}{2} \delta(y + 1) + \frac{3}{8} \mathbf{1}_{(0, 4/3)}(y)$$

The delta term denotes a probability mass, rather than a finite density height.

Output CDF

$$F_Y(y) = \begin{cases} 0, & y < -1, \\ \frac{1}{2}, & -1 \leq y < 0, \\ \frac{1}{2} + \frac{3}{8}y, & 0 \leq y < \frac{4}{3}, \\ 1, & y \geq \frac{4}{3}. \end{cases}$$

The CDF jumps by $1/2$ at -1 , then increases linearly from $(0, 1/2)$ to $(4/3, 1)$.



The constant negative branch creates an atom at $y = -1$; the output never reaches the saturation branch at $y = 2$.

Moments of a quadratic density

Question

A continuous random variable has density

$$f_X(x) = \begin{cases} 3 \times 10^{-3} x^2, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Determine its mean, second raw moment, and variance.

Solution

Mean

$$\mu_X = \int_0^{10} x (3 \times 10^{-3} x^2) dx = 3 \times 10^{-3} \left[\frac{x^4}{4} \right]_0^{10}.$$

$$\mu_X = 7.5.$$

Second raw moment

$$E\{X^2\} = 3 \times 10^{-3} \int_0^{10} x^4 dx = 3 \times 10^{-3} \left[\frac{x^5}{5} \right]_0^{10}.$$

$$E\{X^2\} = 60.$$

Variance and standard deviation

$$\sigma_X^2 = E\{X^2\} - \mu_X^2 = 60 - (7.5)^2.$$

$$\sigma_X^2 = 3.75, \quad \sigma_X = \sqrt{3.75} \approx 1.93649.$$

Moments of an exponential random variable

Question

For $a > 0$, let

$$f_X(x) = ae^{-ax}u(x).$$

Determine the mean, second raw moment, variance, and standard deviation.

Solution

Mean Integration by parts gives

$$\mu_X = a \int_0^{\infty} xe^{-ax} dx = [-xe^{-ax}]_0^{\infty} + \int_0^{\infty} e^{-ax} dx.$$

$$\mu_X = \frac{1}{a}.$$

Second raw moment

$$E\{X^2\} = a \int_0^{\infty} x^2 e^{-ax} dx = [-x^2 e^{-ax}]_0^{\infty} + 2 \int_0^{\infty} x e^{-ax} dx.$$

$$E\{X^2\} = \frac{2}{a^2}.$$

Variance and standard deviation

$$\sigma_X^2 = \frac{2}{a^2} - \left(\frac{1}{a}\right)^2.$$

$$\sigma_X^2 = \frac{1}{a^2}, \quad \sigma_X = \frac{1}{a}.$$

Joint and Conditional Distributions

Joint density and marginal densities

For variables X, Y ,

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y).$$

When a joint density exists,

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y}, \quad f_{X,Y}(x, y) \geq 0,$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1.$$

The support is the region where the density is nonzero. For a region D ,

$$P((X, Y) \in D) = \iint_D f_{X,Y}(x, y) dx dy.$$

Integrate out the other variable to obtain a marginal density:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

Conditional density and moments

For $f_Y(y) > 0$,

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

This is a normalized density in x , with the support restricted to the fixed value y . Similarly, $f_{Y|X}(y | x) = f_{X,Y}(x, y) / f_X(x)$ when $f_X(x) > 0$.

The conditional mean and variance follow from that density:

$$\mu_{X|Y}(y) = E\{X | Y = y\} = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx,$$

$$\sigma_{X|Y}^2(y) = \int_{-\infty}^{\infty} x^2 f_{X|Y}(x | y) dx - \mu_{X|Y}(y)^2.$$

The same integral with x^n gives the conditional moment of order n .

Independence and covariance

Independence means

$$P(X \in A, Y \in B) = P(X \in A) P(Y \in B) .$$

For density-bearing pairs, equivalently,

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) , \quad F_{X,Y}(x, y) = F_X(x) F_Y(y) .$$

Under independence, the conditional density equals its marginal density.

For an integrable function g ,

$$E\{g(X, Y)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy .$$

In particular,

$$\text{Cov}(X, Y) = E\{ (X - \mu_X) (Y - \mu_Y) \} = E\{XY\} - \mu_X \mu_Y .$$

For $\sigma_X, \sigma_Y > 0$, the correlation coefficient is

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} .$$

Independent variables with finite second moments are uncorrelated. Zero covariance does not generally imply independence. It does imply independence for a jointly Gaussian pair; Gaussian marginal densities alone are insufficient.

Example: sine and cosine of a uniform angle

Question

Let Z be uniform on $[0, 2\pi]$, with $X = \sin Z$ and $Y = \cos Z$. Find their marginal densities. Are they uncorrelated? Are they independent?

Solution

For $-1 < x < 1$, the sine map has two inverse branches. Each has derivative magnitude $\sqrt{1-x^2}$ in the forward map, so

$$f_X(x) = \frac{2}{2\pi\sqrt{1-x^2}} = \frac{1}{\pi\sqrt{1-x^2}} .$$

The cosine map gives the same form:

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 < y < 1.$$

Both densities are zero outside their stated intervals. Over one full cycle,

$$\mu_X = \mu_Y = 0,$$

$$E\{XY\} = \frac{1}{2\pi} \int_0^{2\pi} \sin z \cos z \, dz = \frac{1}{4\pi} \int_0^{2\pi} \sin(2z) \, dz = 0.$$

Thus $\text{Cov}(X, Y) = 0$. Nevertheless,

$$X^2 + Y^2 = 1,$$

so knowing X restricts Y : they are dependent. Their joint law lies on a circle and has no ordinary density over the plane.

Example: a joint density on a triangular support

Question

Let

$$f_{X,Y}(x, y) = \begin{cases} Ke^{-x-y}, & x \geq y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find K , the marginal and conditional densities, independence, covariance, and correlation coefficient.

Solution

Normalization gives

$$1 = K \int_0^\infty \int_y^\infty e^{-x-y} \, dx \, dy = \frac{K}{2}, \quad K = 2.$$

The marginal densities are

$$f_X(x) = \int_0^x 2e^{-x-y} \, dy = 2e^{-x} (1 - e^{-x}), \quad x \geq 0,$$

$$f_Y(y) = \int_y^\infty 2e^{-x-y} \, dx = 2e^{-2y}, \quad y \geq 0.$$

Both are zero for negative arguments. The joint support prevents factorization, so the variables are dependent. Conditioning gives

$$f_{X|Y}(x|y) = e^{-(x-y)}, \quad x \geq y, \quad y \geq 0,$$

and zero for $x < y$. With $u = x - y$,

$$\mu_{X|Y}(y) = \int_0^\infty (y+u) e^{-u} du = y+1,$$

$$\sigma_{X|Y}^2(y) = \int_0^\infty u^2 e^{-u} du - 1^2 = 1.$$

Integrating the marginal densities yields

$$\mu_X = \frac{3}{2}, \quad E\{X^2\} = \frac{7}{2}, \quad \sigma_X^2 = \frac{5}{4},$$

$$\mu_Y = \frac{1}{2}, \quad E\{Y^2\} = \frac{1}{2}, \quad \sigma_Y^2 = \frac{1}{4}.$$

Finally,

$$E\{XY\} = \int_0^\infty y(y+1) 2e^{-2y} dy = 1,$$

$$\text{Cov}(X, Y) = 1 - \frac{3}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad \rho_{X,Y} = \frac{1/4}{\sqrt{5/4} \sqrt{1/4}} = \frac{1}{\sqrt{5}}.$$

Worked Problems

Normalization and marginals on a curved support

Question

The joint density is

$$f_{XY}(x, y) = \begin{cases} k(1 + 2xy), & 0 < x \leq 1, \quad 0 < x^2 < y \leq x + 0.5 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find:

- The normalization constant k .
- The marginal densities $f_X(x)$ and $f_Y(y)$.

Solution

Support

$$x + \frac{1}{2} \leq 1 \quad \Rightarrow \quad \mathcal{D} = \{ (x, y) : 0 < x \leq \frac{1}{2}, x^2 < y \leq x + \frac{1}{2} \}.$$

Normalization

$$\begin{aligned} 1 &= k \int_0^{1/2} \int_{x^2}^{x+1/2} (1 + 2xy) \, dy \, dx \\ &= k \int_0^{1/2} \left(\frac{1}{2} + \frac{5}{4}x + x^3 - x^5 \right) dx = k \frac{161}{384}. \end{aligned}$$

$$k = \frac{384}{161}$$

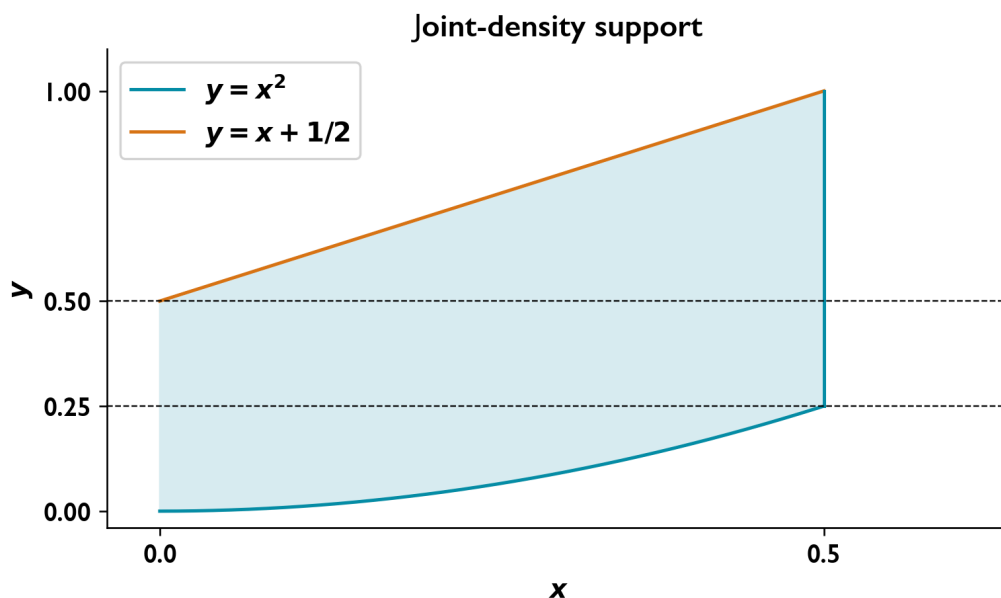
Marginal of X

$$f_X(x) = \begin{cases} \frac{384}{161} \left(\frac{1}{2} + \frac{5}{4}x + x^3 - x^5 \right), & 0 < x < \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Marginal of Y For fixed y , let $\ell = \max(0, y - 1/2)$ and $u = \min(1/2, \sqrt{y})$.

$$f_Y(y) = k \int_{\ell}^u (1 + 2xy) \, dx = k [x + yx^2]_{\ell}^u.$$

$$f_Y(y) = \frac{384}{161} \begin{cases} \sqrt{y} + y^2, & 0 < y < \frac{1}{4}, \\ \frac{1}{2} + \frac{1}{4}y, & \frac{1}{4} \leq y < \frac{1}{2}, \\ 1 - y + y^2 - y^3, & \frac{1}{2} \leq y < 1, \\ 0, & \text{otherwise.} \end{cases}$$



The curved support fixes the marginal integration limits. Horizontal boundary changes occur at $y = 1/4$ and $y = 1/2$.

A joint density on a triangular strip

Question

The joint density is

$$f_{XY}(x, y) = \begin{cases} kx, & 1 \leq x + y \leq 2, \quad x, y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find k , $f_X(x)$, and $f_Y(y)$.

Solution

Normalization Set $s = x + y$; the Jacobian is one.

$$1 = k \int_1^2 \int_0^s x \, dx \, ds = \frac{k}{2} \int_1^2 s^2 \, ds = \frac{7k}{6}, \quad k = \frac{6}{7}.$$

Marginal of X

$$f_X(x) = kx \int_{\max(0, 1-x)}^{2-x} dy.$$

$$f_X(x) = \begin{cases} \frac{6}{7}x, & 0 \leq x \leq 1, \\ \frac{6}{7}x(2-x), & 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Marginal of Y

$$f_Y(y) = k \int_{\max(0, 1-y)}^{2-y} x \, dx.$$

$$f_Y(y) = \begin{cases} \frac{3}{7}(3-2y), & 0 \leq y \leq 1, \\ \frac{3}{7}(2-y)^2, & 1 < y \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

A joint exponential density on an ordered support

Question

Two continuous random variables have joint density

$$f_{X,Y}(x,y) = \begin{cases} ke^{-(x+y)}, & x > y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

- Determine k and both marginal densities.
- Determine whether X and Y are independent.
- Obtain $f_{X|Y}(x|y)$.

Solution

Normalization

$$1 = \int_0^\infty \int_y^\infty ke^{-(x+y)} dx dy = k \int_0^\infty e^{-2y} dy = \frac{k}{2}.$$

$$k = 2.$$

Marginal densities For fixed $x > 0$, the support requires $0 < y < x$:

$$f_X(x) = \int_0^x 2e^{-x-y} dy = 2e^{-x} (1 - e^{-x}).$$

For fixed $y > 0$, the support requires $x > y$:

$$f_Y(y) = \int_y^\infty 2e^{-x-y} dx = 2e^{-2y}.$$

$$f_X(x) = 2e^{-x} (1 - e^{-x}) u(x), \quad f_Y(y) = 2e^{-2y} u(y).$$

Independence For $0 < x < y$, both marginals are positive but the joint density is zero. Therefore it cannot factor as $f_X(x) f_Y(y)$.

X and Y are dependent.

Conditional density For $y > 0$,

$$f_{X|Y}(x|y) = \frac{2e^{-x-y}}{2e^{-2y}} = e^{-(x-y)}, \quad x > y.$$

$$f_{X|Y}(x|y) = \begin{cases} e^{-(x-y)}, & x > y, \\ 0, & x \leq y. \end{cases} \quad y > 0.$$

Its integral over (y, ∞) equals one.

An exponential conditional law with a random rate

Question

Let

$$f_{X,Y}(x,y) = ye^{-y(x+1)} u(x) u(y).$$

Obtain $f_Y(y)$, determine whether the variables are independent, and find $f_{X|Y}(x|y)$.

Solution

Marginal density of Y For $y > 0$,

$$f_Y(y) = \int_0^\infty ye^{-y(x+1)} dx = ye^{-y} \int_0^\infty e^{-yx} dx = e^{-y}.$$

$$f_Y(y) = e^{-y} u(y).$$

Independence For $x \geq 0$,

$$f_X(x) = \int_0^\infty ye^{-(x+1)y} dy = \frac{1}{(x+1)^2}.$$

On the positive quadrant,

$$f_X(x) f_Y(y) = \frac{e^{-y}}{(x+1)^2} \neq ye^{-y(x+1)}.$$

X and Y are dependent.

Conditional density For $y > 0$,

$$f_{X|Y}(x|y) = \frac{ye^{-y(x+1)}}{e^{-y}} = ye^{-yx}, \quad x \geq 0.$$

$$f_{X|Y}(x|y) = \begin{cases} ye^{-yx}, & x \geq 0, \\ 0, & x < 0. \end{cases} \quad y > 0.$$

Normalization follows from $\int_0^\infty ye^{-yx} dx = 1$. The conditional rate depends on y .

Conditional statistics on a curved joint support

Question

Let X and Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} 6xy, & 0 \leq x \leq 1, \quad 0 \leq y \leq \sqrt{x}, \\ 0, & \text{otherwise.} \end{cases}$$

Determine:

- Both marginal densities and whether X and Y are independent.
- The conditional density $f_{X|Y}(x|y)$.
- The conditional mean and variance of X given $Y=y$.
- The joint support in the (x,y) plane and the cross moment $E\{XY\}$.

Solution

Support and normalization

The same region can be written as

$$\mathcal{D} = \{ (x,y) : 0 \leq y \leq 1, y^2 \leq x \leq 1 \}.$$

The support lies between $y = 0$ and $y = \sqrt{x}$ for $0 \leq x \leq 1$:

$$\int_0^1 \int_0^{\sqrt{x}} 6xy \, dy \, dx = \int_0^1 3x^2 \, dx = 1.$$

Marginal densities

$$\begin{aligned} f_X(x) &= \int_0^{\sqrt{x}} 6xy \, dy = 3x^2, & 0 \leq x \leq 1, \\ f_Y(y) &= \int_{y^2}^1 6xy \, dx = 3y(1 - y^4), & 0 \leq y \leq 1. \end{aligned}$$

Both densities are zero outside their stated supports. Thus

$$f_X(x) = 3x^2 \mathbf{1}_{[0,1]}(x), \quad f_Y(y) = 3y(1 - y^4) \mathbf{1}_{[0,1]}(y).$$

The joint density is zero for $x < y^2$, whereas $f_X(x)f_Y(y)$ is positive at interior points in that region. Therefore X and Y are dependent.

Conditional density

For $0 < y < 1$,

$$f_{X|Y}(x|y) = \begin{cases} \frac{2x}{1-y^4}, & y^2 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Conditional moments

$$\begin{aligned} E\{X | Y=y\} &= \frac{2}{1-y^4} \int_{y^2}^1 x^2 dx = \frac{2(1-y^6)}{3(1-y^4)}, \\ E\{X^2 | Y=y\} &= \frac{2}{1-y^4} \int_{y^2}^1 x^3 dx = \frac{1+y^4}{2}. \end{aligned}$$

Hence

$$E\{X | Y=y\} = \frac{2(1+y^2+y^4)}{3(1+y^2)},$$

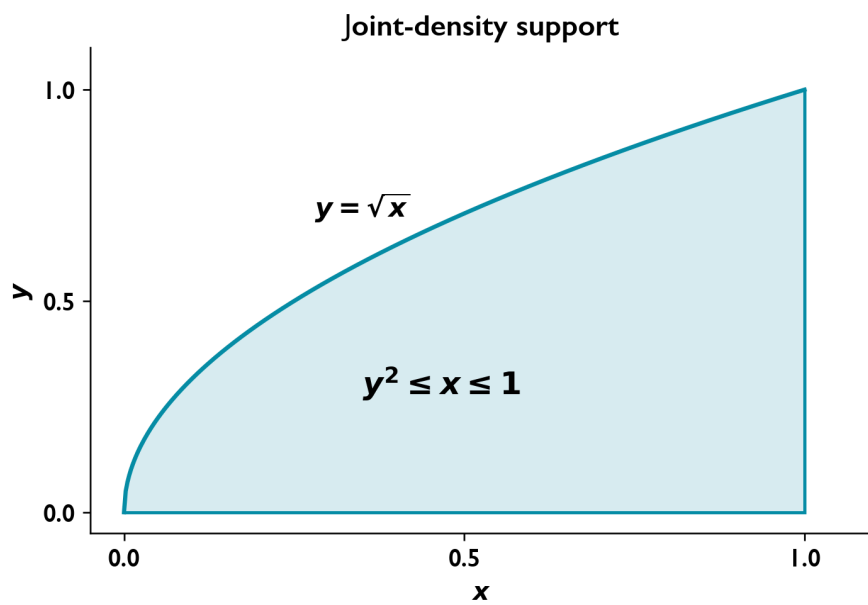
$$\text{Var}(X | Y=y) = \frac{1+y^4}{2} - \left[\frac{2(1+y^2+y^4)}{3(1+y^2)} \right]^2.$$

The density ratio is defined for $0 < y < 1$. Its continuous limits give mean $2/3$ and variance $1/18$ as $y \rightarrow 0$, and mean **1** and variance **0** as $y \rightarrow 1$.

Cross moment

$$\begin{aligned} E\{XY\} &= \int_0^1 \int_0^{\sqrt{x}} 6x^2 y^2 dy dx \\ &= 2 \int_0^1 x^{7/2} dx = \frac{4}{9}. \end{aligned}$$

These are scalar random variables; their joint density determines $E\{XY\}$, not a temporal correlation at an unspecified lag.



The joint support is bounded by $y = \text{sqrt}(x)$, $y = 0$ and $x = 1$.

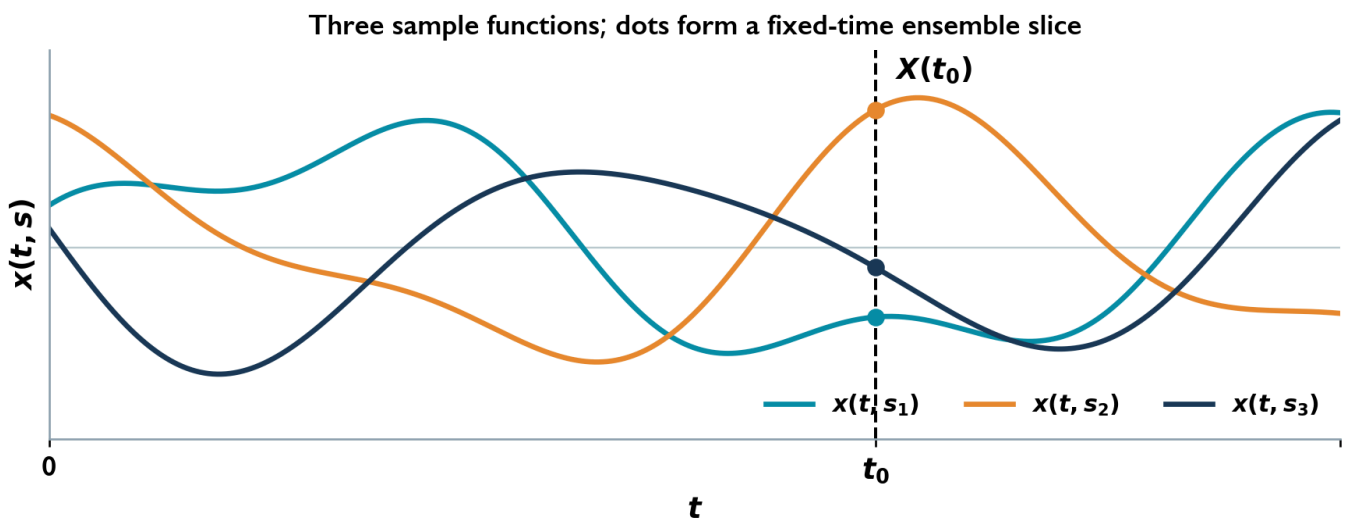
Random processes, stationarity and ergodicity

The process and its ensemble

A random process assigns a waveform to each outcome s of an experiment:

$$X(t, s), \quad x_k(t) = X(t, s_k).$$

Here t is time, and $x_k(t)$ is one sample function, or realization. The collection of possible realizations is the ensemble. Fixing the outcome gives a waveform; fixing the time gives a random variable $X(t_0)$.



The ensemble mean is a deterministic function of time. If a density exists,

$$\mu_X(t) = E\{X(t)\} = \int_{-\infty}^{\infty} x f_{X(t)}(x) dx.$$

Different realizations need not be independent merely because they belong to the same ensemble. Independence is an additional assumption about how observations or experiments are obtained.

What stationarity requires

First-order stationarity means that the one-time probability law is unchanged by a time shift h :

$$F_{X(t+h)}(x) = F_{X(t)}(x) \quad \text{for every } t, h, x.$$

When the mean exists, it is therefore constant. A constant mean alone does not establish this equality.

Strict-sense stationarity requires the joint law of every finite collection $X(t_1), \dots, X(t_n)$ to remain unchanged when all times are shifted by the same h . It concerns all joint probability laws, rather than only a one-time density.

For a real process with finite second moments, wide-sense stationarity (WSS) requires

$$E\{X(t)\} = \mu_X = \text{constant}, \quad R_X(t_1, t_2) = E\{X(t_1)X(t_2)\} = R_X(t_2 - t_1).$$

Writing $\tau = t_2 - t_1$ gives $R_X(\tau) = E\{X(t)X(t + \tau)\}$. Strict-sense stationarity with finite second moments implies WSS. WSS alone does not establish strict-sense stationarity.

Ensemble averages and time averages

An ensemble average compares realizations at fixed times. A time average follows one realization over an observation duration L :

$$\bar{x}_k = \lim_{L \rightarrow \infty} \frac{1}{L} \int_{-L/2}^{L/2} x_k(t) dt.$$

In the almost-sure interpretation used here, mean ergodicity means that this limit equals μ_X for almost every realization. Correlation ergodicity means, for the lag under consideration,

$$R_X(\tau) = \lim_{L \rightarrow \infty} \frac{1}{L} \int_{-L/2}^{L/2} x_k(t) x_k(t + \tau) dt.$$

The limits must exist. Ergodicity is specified for the statistic being recovered; WSS by itself does not justify replacing an ensemble expectation by a time average.

With a normalized resistive load, mean square represents average power:

$$P_X = E\{X^2(t)\} = \mu_X^2 + \sigma_X^2.$$

The deterministic mean contributes power μ_X^2 , while the centered fluctuation has power σ_X^2 . Total RMS is $\sqrt{\mu_X^2 + \sigma_X^2}$; fluctuation RMS is σ_X . These powers coincide with DC and AC power, respectively, when the centered process has no random constant component. For a voltage across resistance r , divide the mean-square voltage by r to obtain power in watts.

Example: Random phase

Question

Is $X(t) = A \cos(2\pi f_c t + \Theta)$ stationary when A and $f_c \neq 0$ are constants and $\Theta \sim \mathcal{U}[0, 2\pi)$?

Solution

A time shift replaces θ by $\theta + 2\pi f_c h$, taken modulo 2π . This phase is still uniform. Because the same phase shift applies to every observed time, all finite-dimensional joint laws remain unchanged: the process is strictly stationary.

$$\mu_X(t) = \frac{A}{2\pi} \int_0^{2\pi} \cos(2\pi f_c t + \theta) d\theta = 0.$$

Its correlation, derived below, is $R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_c \tau)$, so it is also WSS. Integrating one sinusoidal realization over a growing interval gives time mean zero and the same time correlation; it is ergodic for these two statistics.

Example: Random amplitude

Question

Is $X(t) = A \cos(2\pi f_c t)$ strictly stationary if $A \sim \mathcal{U}[0, 1]$ and $f_c \neq 0$?

Solution

Since $E\{A\} = 1/2$,

$$\mu_X(t) = \frac{1}{2} \cos(2\pi f_c t).$$

The mean varies with absolute time. The process is neither strictly stationary nor WSS. A random amplitude with fixed phase therefore behaves differently from a fixed amplitude with uniform random phase.

Example: Random frequency

Question

Show that $X(t) = \sin(2\pi Wt)$ is nonstationary when $W \sim \mathcal{U}[0, B]$ and $B > 0$ hertz.

Solution

For $t \neq 0$,

$$\mu_X(t) = \frac{1}{B} \int_0^B \sin(2\pi wt) dw = \frac{1 - \cos(2\pi Bt)}{2\pi Bt}.$$

At $t = 0$, the mean is zero, also the limit of this expression. The mean is not constant, which rules out both forms of stationarity.

Example: A Gaussian amplitude through an integrator

Question

Let $X(t) = A \cos(2\pi f_c t)$, where $A \sim \mathcal{N}(0, \sigma_A^2)$, $\sigma_A^2 > 0$ and $f_c \neq 0$. Find the law of $Y(t_k)$ for

$$Y(t) = \int_0^t X(u) du,$$

and determine whether $Y(t)$ is stationary.

Solution

Direct integration gives

$$Y(t) = \frac{A \sin(2\pi f_c t)}{2\pi f_c}.$$

At a fixed time t_k , define $K_k = \sin(2\pi f_c t_k) / (2\pi f_c)$. Then

$$E\{Y(t_k)\} = 0, \quad \sigma_Y^2(t_k) = K_k^2 \sigma_A^2.$$

For $K_k \neq 0$, the output is Gaussian with density

$$f_{Y(t_k)}(y) = \frac{1}{\sqrt{2\pi} |K_k| \sigma_A} \exp\left(-\frac{y^2}{2K_k^2 \sigma_A^2}\right).$$

If $K_k = 0$, the output is identically zero and its law is a point mass at zero. The variance changes with time, so the process is neither WSS nor strictly stationary, despite its constant zero mean.

Autocorrelation, autocovariance and crosscorrelation

Energy signals and random processes

For a real deterministic signal $x(t)$ of finite energy,

$$\mathcal{E}_x = \int_{-\infty}^{\infty} x^2(t) dt < \infty, \quad R_x(\tau) = \int_{-\infty}^{\infty} x(t) x(t + \tau) dt.$$

The correlation integrates the overlap between the signal and a shifted copy. At zero lag, $R_x(0) = \mathcal{E}_x$: this is energy, not time-averaged power.

For a real random process, use an ensemble expectation:

$$R_X(t_1, t_2) = E\{X(t_1) X(t_2)\}.$$

Only after verifying WSS may it be reduced to

$$R_X(\tau) = E\{X(t) X(t + \tau)\}, \quad R_X(0) = E\{X^2(t)\} = P_X.$$

For a correlation-ergodic process, the corresponding time integral must be divided by its observation duration before taking the infinite-duration limit. The deterministic energy integral has no such normalization.

Removing the mean

Autocovariance measures the correlation of deviations from the mean:

$$\begin{aligned} C_X(t_1, t_2) &= E\{[X(t_1) - \mu_X(t_1)][X(t_2) - \mu_X(t_2)]\} \\ &= R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2). \end{aligned}$$

For WSS,

$$C_X(\tau) = R_X(\tau) - \mu_X^2, \quad C_X(0) = \sigma_X^2.$$

Thus correlation and covariance coincide for a zero-mean process. Values at two times are uncorrelated when their covariance is zero. Independence implies zero covariance when the moments exist; the converse requires extra assumptions, such as joint Gaussianity.

Conditions every autocorrelation must satisfy

For real WSS processes with finite average power,

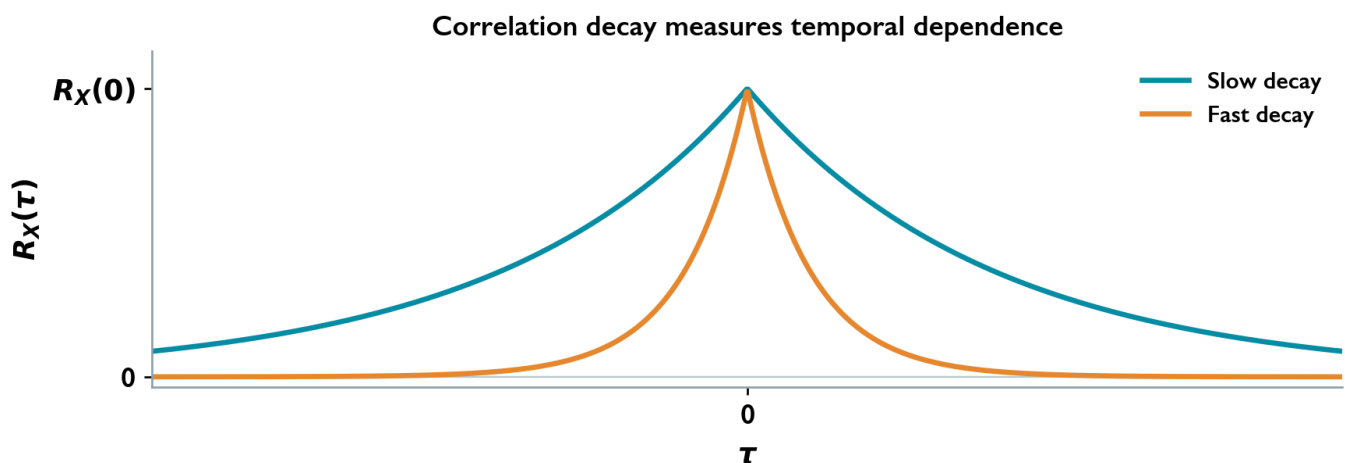
$$R_X(0) \geq 0, \quad R_X(-\tau) = R_X(\tau), \quad |R_X(\tau)| \leq R_X(0).$$

The magnitude bound follows without assuming a positive correlation. Expanding a nonnegative mean square gives

$$0 \leq E\{ [X(t + \tau) \pm X(t)]^2 \} = 2R_X(0) \pm 2R_X(\tau).$$

Hence $-R_X(0) \leq R_X(\tau) \leq R_X(0)$. The same symmetry and magnitude bound hold for finite-energy R_x , with energy at the origin.

These are necessary checks, not a complete validity test. A proposed WSS autocorrelation must also have a nonnegative power spectrum. A negative value of $R_X(\tau)$ at a nonzero lag is allowed; a negative spectral power density is not.



Normalized broad and narrow autocorrelation peaks. The broader peak retains correlation over larger time separations.

A process with slowly varying fluctuations often has a broad correlation peak. Rapid fluctuations can have a narrow peak. This does not imply that every correlation decays monotonically or tends to zero: a sinusoidal component produces a persistent oscillation.

Example: Correlation of a random-phase sinusoid

Question

Find $R_X(\tau)$ for $X(t) = A \cos(2\pi f_c t + \Theta)$ with fixed A, f_c and $\Theta \sim \mathcal{U}[0, 2\pi)$.

Solution

Apply $\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$:

$$R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_c \tau) + \frac{A^2}{2} E\{\cos(4\pi f_c t + 2\pi f_c \tau + 2\Theta)\}.$$

The second term averages to zero over the uniform phase. Therefore

$$R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_c \tau), \quad P_X = \frac{A^2}{2}.$$

It is even, depends only on lag, and attains its greatest magnitude at zero and at other integer half-period lags.

Example: DC and sinusoidal components in noise

Question

How do a deterministic DC offset and an independent random-phase sinusoid affect correlation?

Solution

If $X(t) = a + N(t)$, where a is constant and N is zero-mean WSS noise, expansion gives

$$R_X(\tau) = a^2 + aE\{N(t)\} + aE\{N(t+\tau)\} + R_N(\tau) = a^2 + R_N(\tau).$$

No independence assumption is needed for the constant a . If instead

$X(t) = A \cos(2\pi f_0 t + \Theta) + N(t)$ and the uniform phase is independent of the noise, the cross terms vanish and

$$R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_0 \tau) + R_N(\tau).$$

Example: A finite pulse and a random time shift

Question

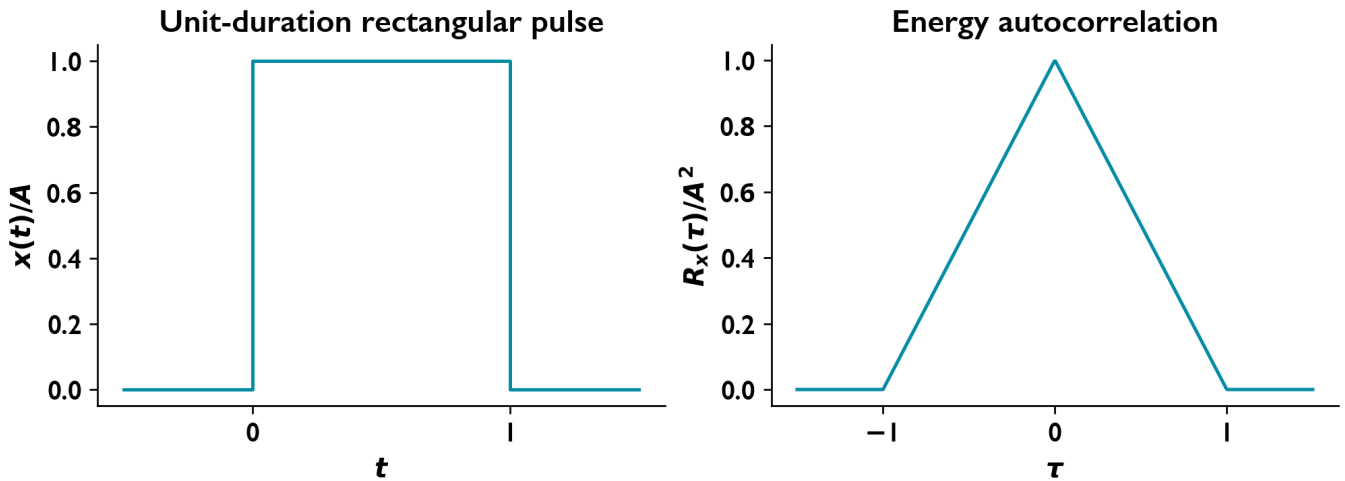
Find the energy correlation of $x(t) = Ap_T(t)$, where $p_T(t) = 1$ on $0 \leq t < T$ and zero elsewhere. Does a single uniformly shifted pulse automatically become WSS?

Solution

The overlap length is $T - |\tau|$ for $|\tau| \leq T$, and zero otherwise:

$$R_x(\tau) = \begin{cases} A^2(T - |\tau|), & |\tau| \leq T, \\ 0, & |\tau| > T. \end{cases}$$

The zero-lag value is A^2T , the pulse energy.



The pulse and its energy correlation for unit duration. The peak is the signal energy.

For a random shift $U \sim \mathcal{U}[0, T]$, take $X(t) = A p_T(t + U)$ with $A \neq 0$. Then $R_X(0, 0) = A^2$ but $R_X(T, T) = 0$. The mean square depends on absolute time, so this single finite pulse is not WSS. Its energy correlation must not be relabeled as a stationary process correlation.

Crosscorrelation and the lag convention

For two real jointly WSS processes, put the lag on the first process:

$$R_{XY}(\tau) = E\{X(t + \tau) Y(t)\}, \quad R_{YX}(\tau) = E\{Y(t + \tau) X(t)\}.$$

Joint WSS means that the individual means are constant and all auto- and crosscorrelations depend only on time differences. Individual WSS alone does not establish this joint property. Reversing the order reflects the lag axis:

$$R_{XY}(\tau) = R_{YX}(-\tau).$$

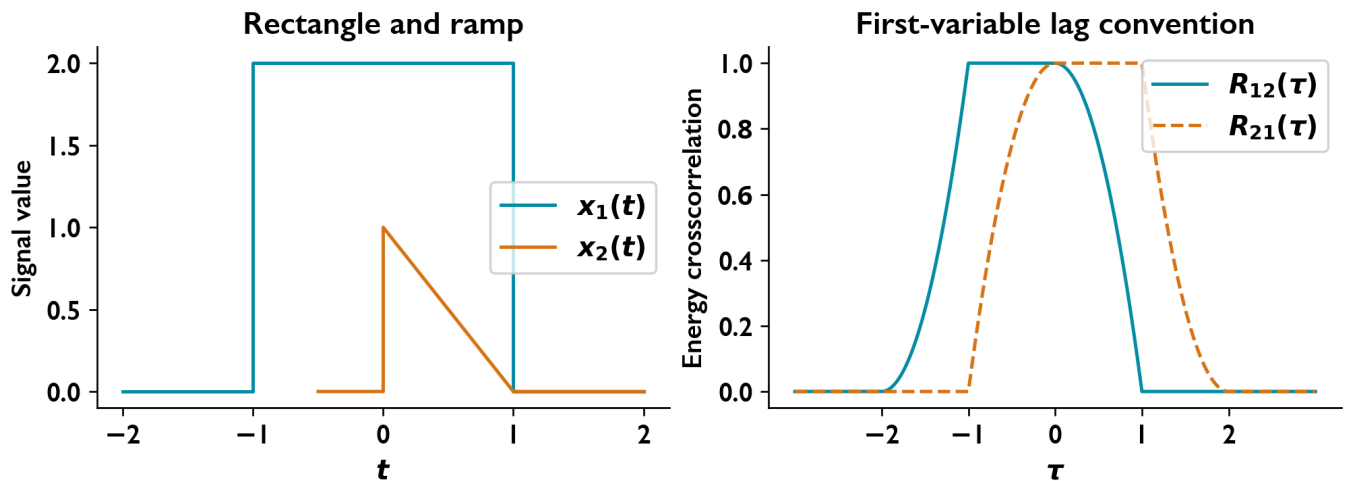
Crosscorrelation need not be even or peak at zero. Crosscovariance is

$$C_{XY}(\tau) = R_{XY}(\tau) - \mu_X \mu_Y.$$

The matching deterministic energy definition is

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t + \tau) y(t) dt,$$

with $R_{xy}(\tau) = R_{yx}(-\tau)$. The order must be fixed before identifying the sign of a delay.



Energy crosscorrelations of a rectangle and ramp. Exchanging the two signals reflects the lag axis.

Worked Problems

Energy autocorrelation of a rectangular pulse

Question

A real deterministic pulse is

$$x(t) = \begin{cases} A, & 0 \leq t \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Obtain and sketch its energy autocorrelation,

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t+\tau)x(t)dt.$$

Solution

Overlap intervals The factors are simultaneously nonzero on the intersection of $[0, 1]$ and $[-\tau, 1-\tau]$.

For $0 \leq \tau \leq 1$,

$$R_x(\tau) = \int_0^{1-\tau} A^2 dt = A^2(1-\tau).$$

For $-1 \leq \tau \leq 0$,

$$R_x(\tau) = \int_{-\tau}^1 A^2 dt = A^2(1+\tau).$$

For $|\tau| > 1$, there is no overlap.

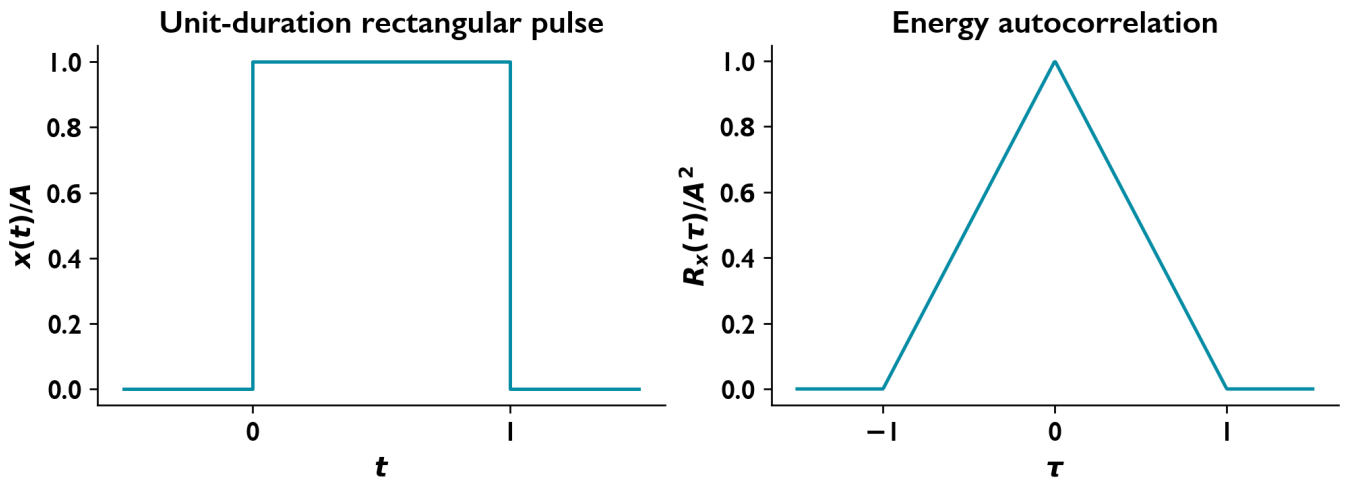
Autocorrelation

$$R_x(\tau) = \begin{cases} A^2(1 - |\tau|), & |\tau| \leq 1, \\ 0, & |\tau| > 1. \end{cases}$$

The graph is triangular and even, with peak A^2 and zeros at $\tau = \pm 1$.

Zero-lag energy

$$R_x(0) = E_x = \int_0^1 A^2 dt = A^2.$$



Overlap of a unit-duration pulse produces its triangular energy autocorrelation.

Crosscorrelation of a rectangular pulse and a ramp

Question

Let

$$x_1(t) = \begin{cases} 2, & -1 \leq t \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad x_2(t) = \begin{cases} 1 - t, & 0 \leq t \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Obtain and sketch the energy crosscorrelations

$$R_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t + \tau) x_2(t) dt,$$

$$R_{21}(\tau) = \int_{-\infty}^{\infty} x_2(t + \tau) x_1(t) dt.$$

Solution

Overlap geometry For R_{12} , intersect the ramp interval $[0, 1]$ with the shifted rectangle interval $[-1 - \tau, 1 - \tau]$. The integrand is $2(1 - t)$ on the intersection.

Partial and full overlaps

$$R_{12}(\tau) = 2 \int_{-1-\tau}^1 (1-t) dt = (\tau+2)^2, \quad -2 \leq \tau < -1,$$

$$R_{12}(\tau) = 2 \int_0^1 (1-t) dt = 1, \quad -1 \leq \tau \leq 0,$$

$$R_{12}(\tau) = 2 \int_0^{1-\tau} (1-t) dt = 1 - \tau^2, \quad 0 < \tau \leq 1.$$

Outside $[-2, 1]$, there is no overlap.

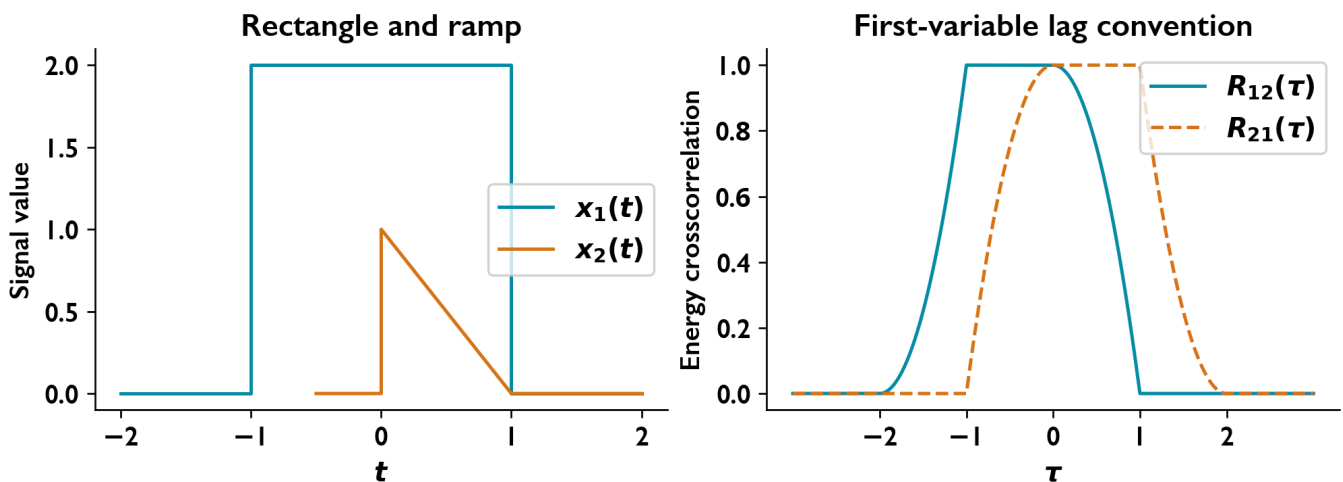
Forward-order crosscorrelation

$$R_{12}(\tau) = \begin{cases} (\tau+2)^2, & -2 \leq \tau < -1, \\ 1, & -1 \leq \tau \leq 0, \\ 1 - \tau^2, & 0 < \tau \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Reverse-order crosscorrelation For real signals, $R_{21}(\tau) = R_{12}(-\tau)$.

$$R_{21}(\tau) = \begin{cases} 1 - \tau^2, & -1 \leq \tau < 0, \\ 1, & 0 \leq \tau \leq 1, \\ (2 - \tau)^2, & 1 < \tau \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

The two graphs are reflections about zero lag. Each has maximum one and continuous joins between adjacent pieces.



Crosscorrelation order is defined by shifting the first signal. Reversal reflects the lag axis.

Testing proposed autocorrelation functions

Question

Determine whether each function can be the autocorrelation of a real WSS process with finite second moment.

- $r_1(\tau) = \sin(2\pi f_0 \tau)$, where $f_0 \neq 0$.
- $r_2(\tau) = \tau^2$.
- The piecewise function

$$r_3(\tau) = \begin{cases} 1 - |\tau|, & |\tau| \leq 1, \\ 1 + |\tau|, & |\tau| > 1. \end{cases}$$

Solution

Necessary conditions A real WSS autocorrelation must satisfy

$$R(-\tau) = R(\tau), \quad |R(\tau)| \leq R(0).$$

Sine candidate r_1 is odd and nonzero, so it is not even. Also $r_1(0) = 0$ while r_1 is nonzero at some lags.

r_1 is invalid.

Quadratic candidate Although r_2 is even,

$$r_2(0) = 0, \quad r_2(1) = 1 > r_2(0).$$

r_2 is invalid.

Piecewise candidate The function is even but violates the zero-lag bound:

$$r_3(0) = 1, \quad r_3(2) = 3 > r_3(0).$$

r_3 is invalid.

Power spectra, linear systems and random binary signals

Finite observation and power spectrum

Observe a process over $[-T, T]$ and define its finite-window transform:

$$\Xi_T(f) = \int_{-T}^T X(t) e^{-j2\pi ft} dt.$$

Its power spectrum is

$$S_X(f) = \lim_{T \rightarrow \infty} \frac{E\{|\Xi_T(f)|^2\}}{2T},$$

when the limit exists. Spectral delta lines are interpreted by their weights.

Correlation and spectrum

For a real WSS process, autocorrelation and the two-sided power spectral density form a Fourier pair:

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau} d\tau,$$

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau} df.$$

Here f is frequency in hertz, τ is time lag, and $j = \sqrt{-1}$. The spectrum distributes mean-square power over frequency:

$$P_X = R_X(0) = \int_{-\infty}^{\infty} S_X(f) df, \quad S_X(f) \geq 0, \quad S_X(-f) = S_X(f).$$

With amplitude in volts, the spectral units are V^2/Hz ; for a normalized load they can be expressed as watts per hertz. When the integral exists, $S_X(0) = \int R_X(\tau) d\tau$.

For a deterministic finite-energy signal, the corresponding Fourier transform of $R_x(\tau)$ is the energy spectral density $\psi_x(f) = |\xi_x(f)|^2$, where $\xi_x(f) = \int x(t) e^{-j2\pi ft} dt$. Its spectral area is energy. It is not the PSD of a WSS random process.

DC and sinusoidal lines

Writing $X(t) = \mu_X + \tilde{X}(t)$ with a zero-mean WSS fluctuation gives

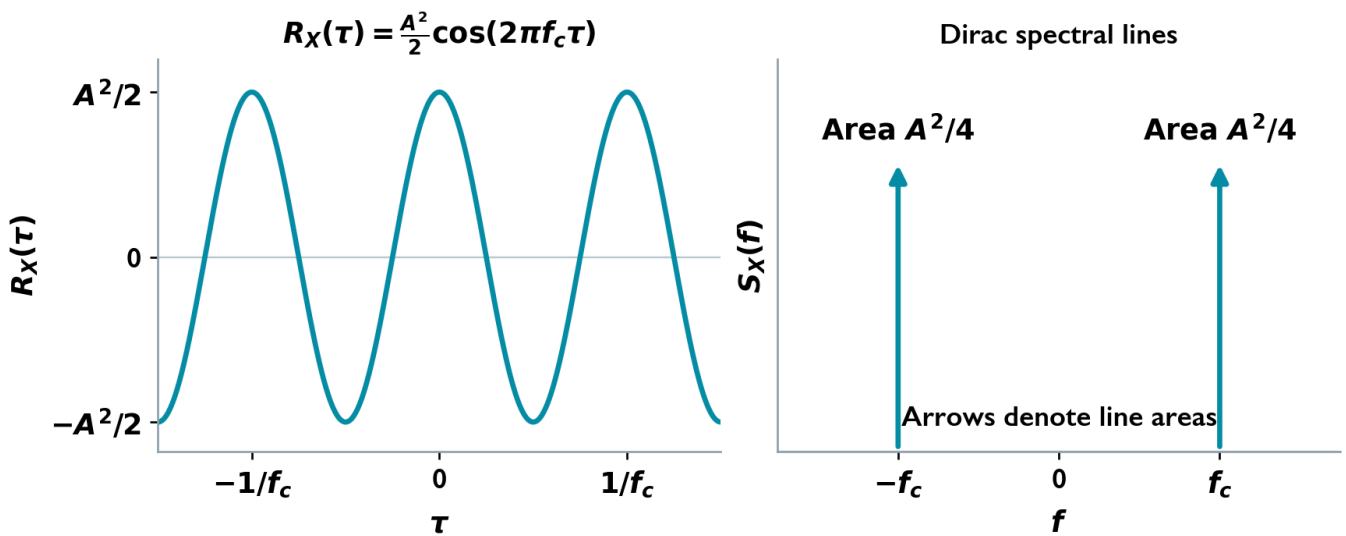
$$R_X(\tau) = \mu_X^2 + C_X(\tau), \quad S_X(f) = \mu_X^2 \delta(f) + S_{\tilde{X}}(f).$$

The deterministic mean contributes a spectral line at zero. A delta line is characterized by its area or weight, not by an ordinary finite height. A DC line can also contain power from a random constant component; its weight alone does not determine the mean.

For the fixed-amplitude random-phase sinusoid,

$$R_X(\tau) = \frac{A^2}{2} \cos(2\pi f_c \tau), \quad S_X(f) = \frac{A^2}{4} [\delta(f - f_c) + \delta(f + f_c)].$$

Each line carries power $A^2/4$, giving total power $A^2/2$.



Cosine correlation and the two spectral lines. Arrow labels indicate delta weights.

Example: Triangular correlation and triangular spectrum

Question

Transform a triangular correlation, and then find the correlation corresponding to a triangular PSD.

Solution

Define the dimensionless functions

$$\Lambda(u) = \begin{cases} 1 - |u|, & |u| \leq 1, \\ 0, & |u| > 1, \end{cases} \quad \text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \quad \text{sinc}(0) = 1.$$

For a zero-mean process and time width $T > 0$,

$$R_X(\tau) = \sigma^2 \Lambda(\tau/T) \quad \Leftrightarrow \quad S_X(f) = \sigma^2 T \text{sinc}^2(fT).$$

For a frequency width $f_0 > 0$ and peak spectral density s_0 ,

$$S_X(f) = s_0 \Lambda(f/f_0) \Leftrightarrow R_X(\tau) = s_0 f_0 \text{sinc}^2(f_0 \tau).$$

The zero-lag values agree with the spectral areas: σ^2 for the first pair and $s_0 f_0$ for the second. A triangle in one domain becomes a squared sinc in the other; the scale factor follows the integration variable.

Example: A triangular PSD with a DC line

Question

For $S_X(f) = s_0 \Lambda(f/f_0) + p_0 \delta(f)$, find correlation, DC and AC powers, and sampling intervals giving uncorrelated samples. Interpret the line as a deterministic mean contribution.

Solution

The inverse transform gives

$$R_X(\tau) = s_0 f_0 \text{sinc}^2(f_0 \tau) + p_0.$$

Thus $\mu_X^2 = p_0$, $P_{\text{DC}} = p_0$, $P_{\text{AC}} = s_0 f_0$, and $P_X = p_0 + s_0 f_0$. The stated numerical unit-height, unit-line case is obtained with $s_0 = 1$ and $p_0 = 1$ in the relevant units.

After subtracting the mean contribution,

$$C_X(\tau) = s_0 f_0 \text{sinc}^2(f_0 \tau).$$

Consequently $C_X(kT_s) = 0$ for every nonzero integer k if

$$T_s = \frac{m}{f_0}, \quad f_s = \frac{f_0}{m}, \quad m = 1, 2, \dots$$

These samples are uncorrelated, although $R_X(kT_s) = p_0$ rather than zero. They need not be independent without joint Gaussianity. If the line were a random constant component instead, this covariance conclusion would require its mean and variance to be specified.

Cross spectra

With $R_{XY}(\tau) = E\{X(t+\tau)Y(t)\}$, define

$$S_{XY}(f) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j2\pi f\tau} d\tau,$$

$$R_{XY}(\tau) = \int_{-\infty}^{\infty} S_{XY}(f) e^{j2\pi f\tau} df.$$

For real jointly WSS processes,

$$S_{XY}(f) = S_{YX}(-f) = S_{YX}^*(f).$$

A cross spectrum describes the frequency relationship between two processes. It may be complex and is not required to be nonnegative. Its integral gives the zero-lag crosscorrelation, $R_{XY}(0) = E\{X(t)Y(t)\}$.

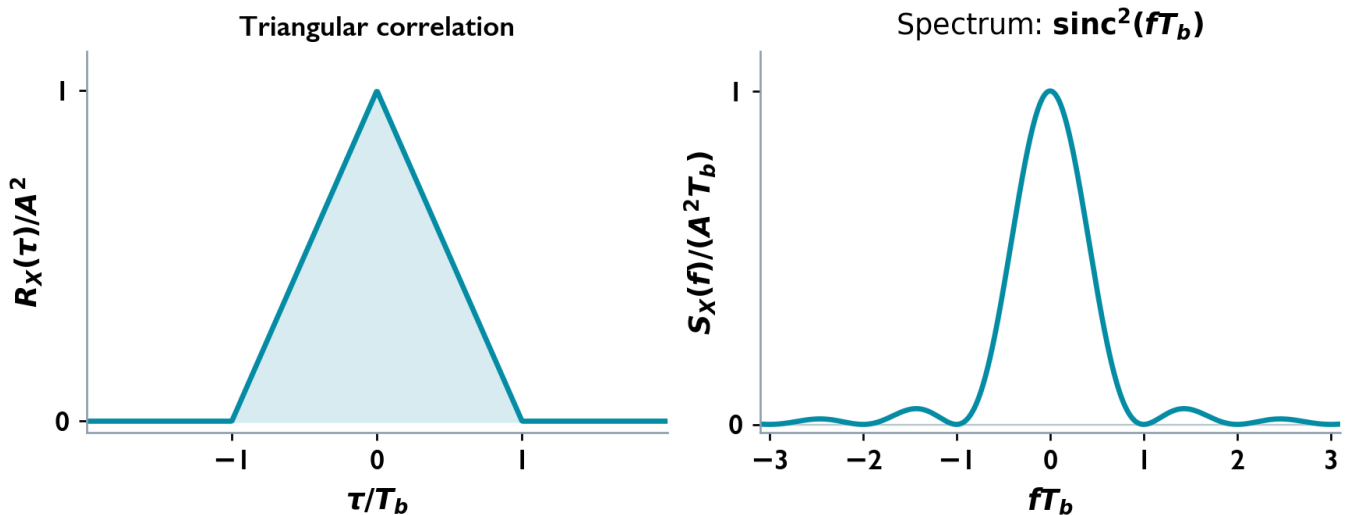
A random bipolar binary sequence

Let independent equiprobable symbols $A_k \in \{-1, +1\}$ be held for duration T . Randomize the timing origin uniformly over $[0, T)$, independently of the symbols, so no absolute time is privileged. Then $E\{X(t)\} = 0$ and $P_X = 1$.

For $|\tau| \leq T$, the probability that two observations belong to the same symbol is $1 - |\tau|/T$. The same-symbol product is one; different symbols have expected product zero. Hence

$$R_X(\tau) = \begin{cases} 1 - |\tau|/T, & |\tau| \leq T, \\ 0, & |\tau| > T, \end{cases} \quad S_X(f) = T \operatorname{sinc}^2(fT).$$

For symbols $\pm A$, both expressions are multiplied by A^2 .



Binary correlation and PSD for two symbol durations, using shared axis scales.

The bit rate is $R_b = 1/T$. First spectral zeros occur at $f = \pm 1/T = \pm R_b$, so the null-to-null main-lobe width is $2R_b$. A longer symbol produces broader correlation and a narrower main lobe; a shorter symbol reverses this relation. The squared-sinc sidelobes continue outside the main lobe, so this width is not a strict finite spectral support.

Worked Problems

Power spectrum of the sum of two processes

Question

Let $X(t)$ and $Y(t)$ be zero-mean WSS processes, and define $Z(t) = X(t) + Y(t)$. Assume joint WSS for the general spectral calculation.

- Determine $S_Z(f)$ in terms of the auto and cross spectra.
- Simplify the result when $X(t)$ and $Y(t)$ are independent.

Solution

Expand the autocorrelation

Using $R_{XY}(\tau) = E\{X(t+\tau)Y(t)\}$,

$$\begin{aligned} R_Z(\tau) &= E\{[X(t+\tau) + Y(t+\tau)][X(t) + Y(t)]\} \\ &= R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau). \end{aligned}$$

Take the Fourier transform

Since $S_{YX}(f) = S_{XY}^*(f)$,

$$S_Z(f) = S_X(f) + S_Y(f) + 2 \operatorname{Re} S_{XY}(f).$$

Independent processes

Independence and zero means give $R_{XY}(\tau) = 0$ at every lag. Therefore,

$$S_Z(f) = S_X(f) + S_Y(f).$$

The same simplification holds for processes with zero cross-correlation at every lag.

A spectral null produced by delayed-symbol subtraction

Question

An i.i.d. sequence $\{\alpha_n\}$ takes values -1 and $+1$ with equal probability. For real K , define

$$\beta_n = \alpha_n + K\alpha_{n-3}, \quad X(t) = \sum_{n=-\infty}^{\infty} \beta_n g(t - nT).$$

Determine K if the power spectrum has a null at $f = 1/(3T)$. Assume $G(1/(3T)) \neq 0$, where $G(f)$ is the Fourier transform of $g(t)$.

Solution

Symbol correlation

The original symbols have zero mean and unit variance, so

$$R_\alpha[k] = \delta[k],$$

where $\delta[k]$ is the Kronecker delta. Expanding the filtered-symbol product gives

$$R_\beta[k] = (1 + K^2) \delta[k] + K\delta[k-3] + K\delta[k+3].$$

Spectral factor

$$\begin{aligned} \Phi_\beta(f) &= \sum_{k=-\infty}^{\infty} R_\beta[k] e^{-j2\pi f k T} \\ &= 1 + K^2 + 2K \cos(6\pi f T). \end{aligned}$$

The fixed-clock pulse train has the period-averaged PSD

$$\bar{S}_X(f) = \frac{|G(f)|^2}{T} \Phi_\beta(f).$$

The same expression is the WSS PSD when an independent uniform timing offset over $[0, T)$ is included.

Apply the null condition

At $f = 1/(3T)$,

$$\Phi_\beta\left(\frac{1}{3T}\right) = 1 + K^2 + 2K = (1 + K)^2.$$

Because the pulse spectrum is nonzero there,

$$(1 + K)^2 = 0 \Rightarrow K = -1.$$

If $G(1/(3T)) = 0$, that null alone would impose no restriction on K .

Random Processes Through Linear Filters

Convolution, mean, and correlation

Let a real WSS process $X(t)$ pass through a real LTI filter with impulse response $h(t)$ and frequency response $H(f)$. For an output with finite second moment,

$$Y(t) = h(t) * X(t) = \int_{-\infty}^{\infty} h(u) X(t-u) du.$$

Taking the expectation inside the integral gives

$$\mu_Y = \int_{-\infty}^{\infty} h(u) \mu_X du = \mu_X H(0).$$

The output mean depends on the filter's DC gain. A zero-mean input therefore gives a zero-mean output.

Expanding the expected product of two output values gives

$$\begin{aligned} R_Y(\tau) &= E\{Y(t+\tau) Y(t)\} \\ &= \iint h(u) h(v) R_X(\tau-u+v) du dv \\ &= R_X(\tau) * h(\tau) * h(-\tau). \end{aligned}$$

The mean is constant and the correlation depends only on lag, so the output is WSS under the stated existence condition. For Gaussian input, the output is also Gaussian.

Power spectra and input-output correlation

The Fourier transform converts the correlation convolution into

$$S_Y(f) = |H(f)|^2 S_X(f).$$

The squared magnitude is the power gain. The output mean square and variance are

$$P_Y = R_Y(0) = \int_{-\infty}^{\infty} |H(f)|^2 S_X(f) df, \quad \sigma_Y^2 = P_Y - \mu_Y^2.$$

Use the cross-correlation convention

$$R_{XY}(\tau) = E\{X(t+\tau) Y(t)\}.$$

Then

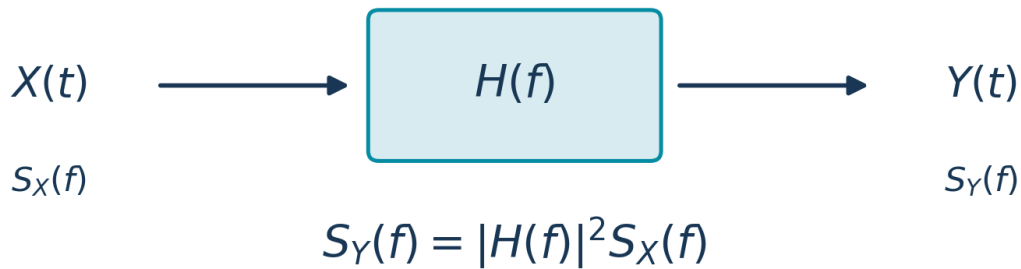
$$R_{XY}(\tau) = \int h(u) R_X(\tau + u) du = R_X(\tau) * h(-\tau),$$

$$R_{YX}(\tau) = R_{XY}(-\tau) = R_X(\tau) * h(\tau).$$

Their cross spectra are

$$S_{XY}(f) = S_X(f) H^*(f), \quad S_{YX}(f) = S_X(f) H(f).$$

The output PSD retains the magnitude of the filter response; the cross spectrum also retains its phase.



Input and output power spectra of an LTI filter.

Example: Cascaded filters

Question

A WSS process passes through two real filters: $X \rightarrow H_1 \rightarrow V \rightarrow H_2 \rightarrow Y$. Find the intermediate and final spectra and correlations.

Solution

The equivalent response is

$$H_{\text{eq}}(f) = H_1(f) H_2(f), \quad h_{\text{eq}} = h_1 * h_2.$$

Consequently,

$$S_V(f) = |H_1(f)|^2 S_X(f),$$

$$S_Y(f) = |H_2(f)|^2 S_V(f) = |H_1(f) H_2(f)|^2 S_X(f).$$

In the time domain,

$$R_V = R_X * h_1(\tau) * h_1(-\tau),$$

$$R_Y = R_X * h_{\text{eq}}(\tau) * h_{\text{eq}}(-\tau).$$

The intermediate process remains the output of the first filter; it is not a new independent input.

Example: A random-phase sinusoid through an RC filter

Question

Let $X(t) = A \cos(2\pi f_0 t + \Theta)$, where Θ is uniform on $[0, 2\pi)$, and let

$$H(f) = \frac{1}{1 + j2\pi RCf}.$$

Find the output autocorrelation and power.

Solution

The input has zero mean and

$$S_X(f) = \frac{A^2}{4} [\delta(f - f_0) + \delta(f + f_0)].$$

Evaluate the power gain at the two spectral lines:

$$S_Y(f) = \frac{A^2}{4 [1 + (2\pi RCf_0)^2]} [\delta(f - f_0) + \delta(f + f_0)].$$

Inverse transformation gives

$$R_Y(\tau) = \frac{A^2}{2 [1 + (2\pi RCf_0)^2]} \cos(2\pi f_0 \tau).$$

Thus

$$P_Y = R_Y(0) = \frac{A^2}{2 [1 + (2\pi RCf_0)^2]}.$$

The filter changes amplitude and phase, while the autocorrelation reflects the amplitude change.

Worked Problems

Independent Gaussian noise branches and an estimation-error

network

Question

Independent Gaussian white-noise processes $W_1(t)$ and $W_2(t)$ satisfy

$$R_{W_i}(\tau) = \frac{\eta}{2} \delta(\tau).$$

Identical real LTI filters $h(t)$ produce $N_1(t)$ and $N_2(t)$. Define

$$N = N_1 - N_2, \quad \varepsilon = N_1 - h_o * N.$$

Find:

- The density of N and the joint density of N_1, N_2 at a fixed time.
- $P(N > -1)$.
- A simplified expression for ε .

Solution

Branch statistics Assume $\eta > 0$ and $0 < E_h < \infty$.

$$E_h = \int_{-\infty}^{\infty} |h(u)|^2 du, \quad \sigma_b^2 = \frac{\eta E_h}{2}, \quad S_{N_i}(f) = \frac{\eta}{2} |H(f)|^2.$$

The branch outputs are independent, zero-mean Gaussian processes.

Densities and probability

$$N \sim \mathcal{N}(0, \eta E_h).$$

$$f_N(n) = \frac{e^{-n^2/(2\eta E_h)}}{\sqrt{2\pi\eta E_h}}$$

$$f_{N_1 N_2}(n_1, n_2) = \frac{1}{2\pi\sigma_b^2} \exp\left[-\frac{n_1^2 + n_2^2}{2\sigma_b^2}\right]$$

$$P(N > -1) = 1 - Q\left(\frac{1}{\sqrt{\eta E_h}}\right)$$

A numerical probability requires ηE_h .

Error network

$$\varepsilon = (\delta - h_o) * N_1 + h_o * N_2$$

$$S_\varepsilon(f) = \frac{\eta}{2} |H(f)|^2 (|1 - H_o(f)|^2 + |H_o(f)|^2).$$

Without a specified H_o , this is the general result. For minimum mean-square estimation,

$$H_o = \frac{S_{N_1 N}}{S_N} = \frac{1}{2}, \quad h_o = \frac{1}{2}\delta,$$

$$\varepsilon = \frac{N_1 + N_2}{2}, \quad \text{Var}(\varepsilon) = \frac{\eta E_h}{4}$$

If $\eta E_h = 0$, the outputs vanish in mean square, the joint law is a point mass at the origin, and $P(N > -1) = 1$.

Output autocorrelation of a rectangular LTI filter

Question

A WSS input has

$$R_X(\tau) = \begin{cases} 1 - |\tau|, & |\tau| \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $h(t) = u(t) - u(t-1)$, use convolution to find $R_Y(\tau)$.

Solution

Correlation convolution

$$\begin{aligned} R_Y &= R_X * h * \tilde{h}, & \tilde{h}(t) &= h(-t), \\ h * \tilde{h} &= \Lambda, & \Lambda(\tau) &= \max(1 - |\tau|, 0), \\ R_Y &= \Lambda * \Lambda. \end{aligned}$$

Overlap integral

$$R_Y(\tau) = \int_{-1}^1 (1 - |v|) \max(1 - |\tau - v|, 0) dv.$$

Set $a = |\tau|$ and split at the changes of sign inside the overlap:

$$R_Y(\tau) = \begin{cases} \frac{2}{3} - a^2 + \frac{1}{2}a^3, & 0 \leq a \leq 1, \\ \frac{1}{6}(2 - a)^3, & 1 < a \leq 2, \\ 0, & a > 2, \end{cases} \quad a = |\tau|.$$

$$P_Y = R_Y(0) = \frac{2}{3}$$

At $|\tau| = 1$, both branches equal $1/6$.

Autocorrelation and spectrum of a shifted difference

Question

A real WSS process $X(t)$ is transformed according to

$$Y(t) = X(t+a) - X(t-a).$$

Determine:

- The autocorrelation $R_Y(\tau)$ in terms of $R_X(\tau)$.
- The PSD $S_Y(f)$ in terms of $S_X(f)$.

Solution

Expand the autocorrelation

The output mean is zero. With $R_X(\tau) = E\{X(t+\tau)X(t)\}$,

$$\begin{aligned} R_Y(\tau) &= E\{[X(t+\tau+a) - X(t+\tau-a)][X(t+a) - X(t-a)]\} \\ &= 2R_X(\tau) - R_X(\tau-2a) - R_X(\tau+2a). \end{aligned}$$

Thus

$$R_Y(\tau) = 2R_X(\tau) - R_X(\tau-2a) - R_X(\tau+2a).$$

Find the transfer function

$$H(f) = e^{j2\pi fa} - e^{-j2\pi fa} = 2j \sin(2\pi fa).$$

Consequently,

$$S_Y(f) = 4 \sin^2(2\pi fa) S_X(f).$$

Zero-lag check

Evenness of R_X gives

$$P_Y = R_Y(0) = 2[R_X(0) - R_X(2a)].$$

White Noise, Filtered Noise, and Equivalent Bandwidth

The white-noise model

Zero-mean ideal white noise has a constant two-sided power spectral density:

$$S_W(f) = \frac{N_0}{2}, \quad -\infty < f < \infty.$$

Its autocorrelation is the inverse Fourier transform:

$$R_W(\tau) = \frac{N_0}{2} \delta(\tau).$$

The model has no correlation at nonzero lag. Its total ideal power is infinite, so finite noise power is evaluated after bandwidth limitation rather than by assigning a finite variance to an ideal unfiltered point value. Whiteness specifies the spectrum; Gaussianity specifies the probability law.

When the density represents available thermal-noise power,

$$N_0 = kT, \quad S_W(f) = \frac{kT}{2} \text{ W/Hz},$$

where $k \approx 1.38 \times 10^{-23}$ J/K and T is absolute temperature in kelvin. The negative and positive frequencies each contribute half the available power. An ideal low-pass bandwidth B therefore admits kTB . This available-power convention does not by itself specify a resistor's voltage-noise density.

Filtering the spectrum

For white noise passing through a filter,

$$S_Y(f) = \frac{N_0}{2} |H(f)|^2,$$

$$P_Y = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df.$$

The integral must be finite for finite output noise power. The filter shapes the flat input spectrum into colored noise. Gaussian white input gives Gaussian output.

Example: Ideal low-pass filtering

Question

White Gaussian noise with PSD $N_0/2$ passes through a unity-gain ideal filter with passband $[-B, B]$, where $B > 0$. Find its output PSD, autocorrelation, and power.

Solution

The output PSD is

$$S_Y(f) = \begin{cases} N_0/2, & |f| \leq B, \\ 0, & |f| > B. \end{cases}$$

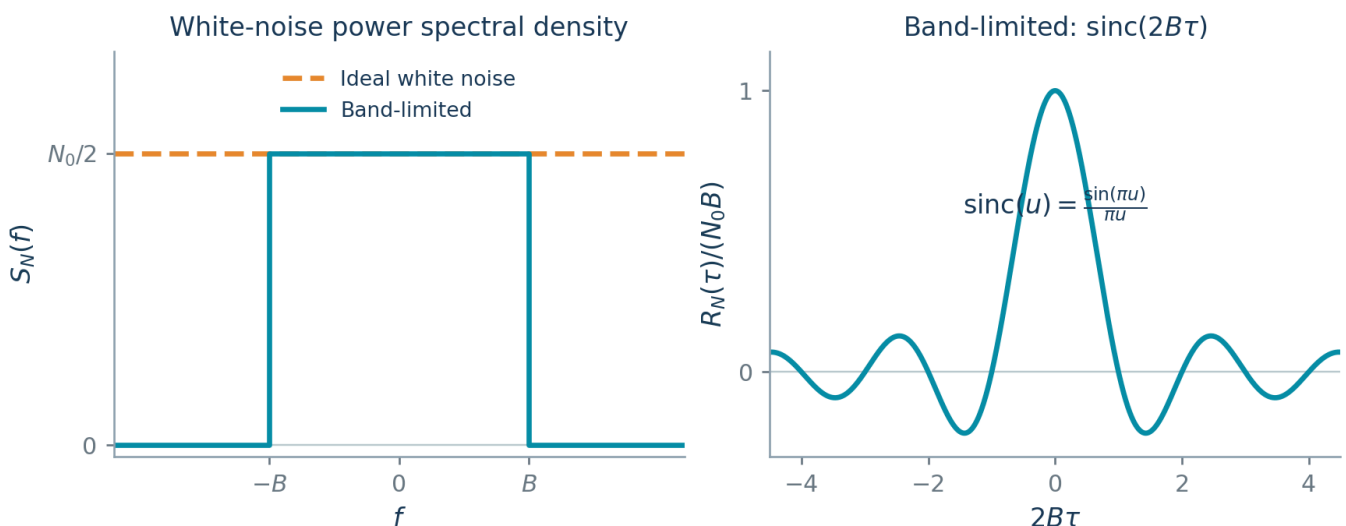
Using $\text{sinc}(u) = \sin(\pi u) / (\pi u)$, with $\text{sinc}(0) = 1$,

$$\begin{aligned} R_Y(\tau) &= \frac{N_0}{2} \int_{-B}^B e^{j2\pi f\tau} df \\ &= N_0 B \text{sinc}(2B\tau). \end{aligned}$$

Therefore,

$$P_Y = \sigma_Y^2 = N_0 B.$$

Its correlation zeros occur at $\tau = m / (2B)$ for nonzero integers m . Distinct Gaussian samples on that grid are independent; at other separations, they are generally correlated.



White-noise PSD and the finite-bandwidth output correlation.

Example: RC filtering

Question

White Gaussian noise passes through an RC low-pass filter. Find the output spectrum, power, and autocorrelation.

Solution

For $RC > 0$,

$$H(f) = \frac{1}{1 + j2\pi RCf}, \quad f_{3\text{dB}} = \frac{1}{2\pi RC}.$$

Hence

$$S_Y(f) = \frac{N_0/2}{1 + (2\pi RCf)^2}.$$

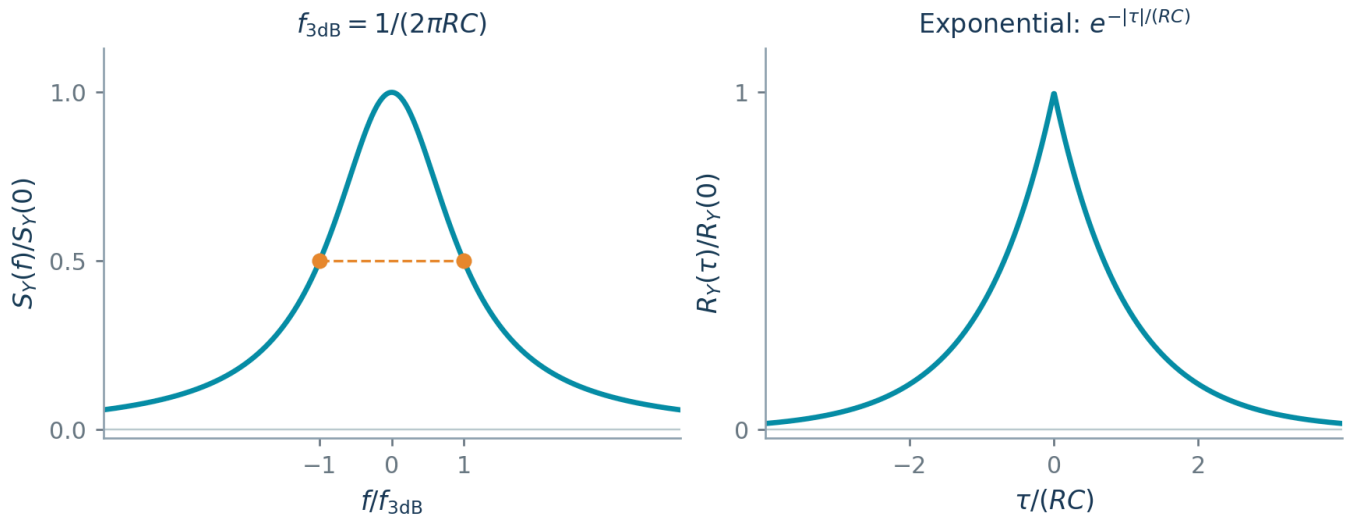
Substitute $u = 2\pi RCf$ in the power integral:

$$\begin{aligned} P_Y &= N_0 \int_0^\infty \frac{df}{1 + (2\pi RCf)^2} \\ &= \frac{N_0}{2\pi RC} \int_0^\infty \frac{du}{1 + u^2} \\ &= \frac{N_0}{4RC} = \frac{\pi}{2} N_0 f_{3\text{dB}}. \end{aligned}$$

The corresponding correlation is

$$R_Y(\tau) = \frac{N_0}{4RC} e^{-|\tau|/(RC)}.$$

Its zero-lag value confirms the same power. Increasing RC lowers the noise power and broadens the correlation in time.



RC noise spectrum and exponential correlation.

Noise-equivalent bandwidth

Let $g = \max_f |H(f)|^2 > 0$ be the reference power gain of a real filter. Define its one-sided noise-equivalent bandwidth by

$$B_N = \frac{1}{g} \int_0^\infty |H(f)|^2 df = \frac{1}{2g} \int_{-\infty}^\infty |H(f)|^2 df.$$

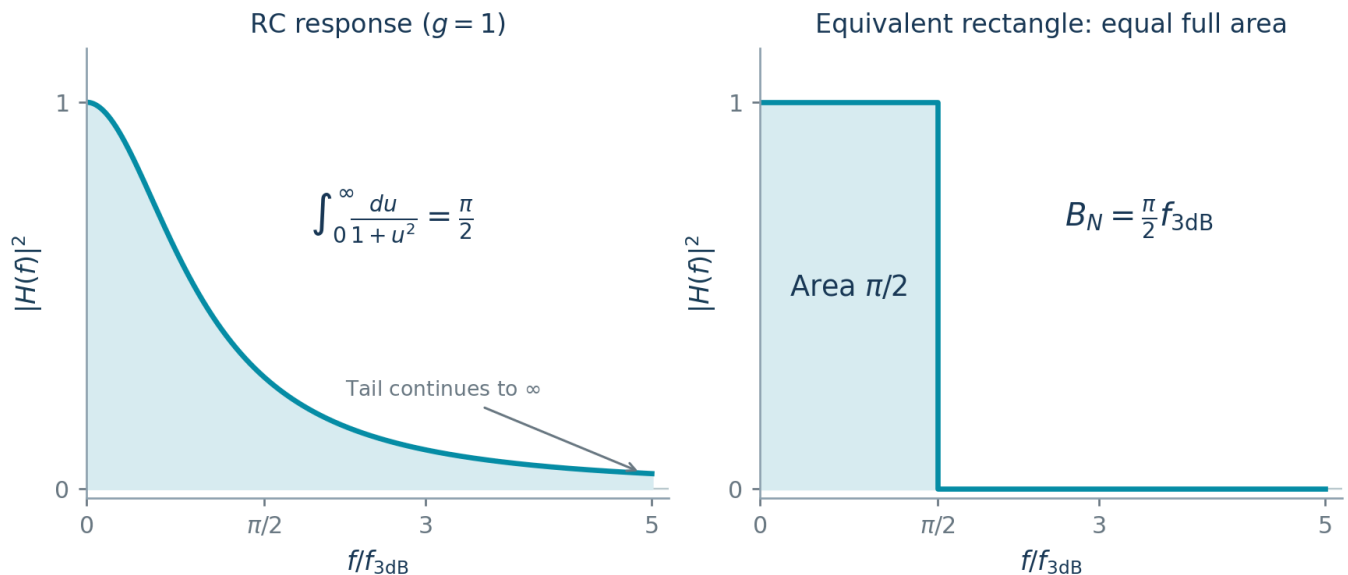
An ideal rectangle of height g and positive-frequency width B_N has the same area as the squared-gain curve. Therefore,

$$P_Y = N_0 g B_N.$$

For the RC filter, $g = 1$, giving

$$B_N = \int_0^\infty \frac{df}{1 + (2\pi RCf)^2} = \frac{1}{4RC} = \frac{\pi}{2} f_{3dB}.$$

The 3-dB frequency marks one point on the response; equivalent bandwidth measures its area.



Equal positive-frequency areas define the equivalent bandwidth.

Worked Problems

White noise through a first-order low-pass filter

Question

Zero-mean white noise $N_i(t)$ has

$$R_{N_i}(\tau) = \frac{N_0}{2} \delta(\tau).$$

It drives a filter with

$$H(\omega) = \frac{1}{1 + j\omega/\omega_0}, \quad \omega_0 > 0, \quad \omega = 2\pi f.$$

Determine:

- The impulse response.
- The output-noise autocorrelation.
- Both orders of input-output cross-correlation.
- The output noise power.

Solution

Impulse response

$$H(\omega) = \frac{\omega_0}{\omega_0 + j\omega} \Rightarrow h(t) = \omega_0 e^{-\omega_0 t} u(t).$$

Output spectrum and autocorrelation

$$S_{N_o}(f) = \frac{N_0}{2} |H(f)|^2$$

$$= \frac{N_0 \omega_0^2}{2 [\omega_0^2 + (2\pi f)^2]}.$$

Using $e^{-a|\tau|} \leftrightarrow 2a / [a^2 + (2\pi f)^2]$,

$$R_{N_o}(\tau) = \frac{N_0 \omega_0}{4} e^{-\omega_0 |\tau|}.$$

Input-output cross-correlations

With $R_{XY}(\tau) = E\{X(t + \tau)Y(t)\}$,

$$R_{N_i N_o}(\tau) = \int_{-\infty}^{\infty} h(v) R_{N_i}(\tau + v) dv$$

$$= \frac{N_0}{2} h(-\tau).$$

Therefore,

$$R_{N_i N_o}(\tau) = \frac{N_0 \omega_0}{2} e^{\omega_0 \tau} u(-\tau),$$

$$R_{N_o N_i}(\tau) = \frac{N_0 \omega_0}{2} e^{-\omega_0 \tau} u(\tau).$$

Output noise power

The output mean is zero, so

$$P_{N_o} = R_{N_o}(0) = \frac{N_0 \omega_0}{4}.$$

Autocorrelation of ideal low-pass filtered white noise

Question

White noise has two-sided PSD $N_0/2$ and passes through

$$H(f) = \begin{cases} 1, & |f| \leq B, \\ 0, & \text{otherwise,} \end{cases} \quad B > 0.$$

Determine the output-noise autocorrelation.

Solution

Output spectrum

$$S_{N_o}(f) = \frac{N_0}{2} \text{rect}\left(\frac{f}{2B}\right).$$

Inverse Fourier transform

$$\begin{aligned} R_{N_o}(\tau) &= \frac{N_0}{2} \int_{-B}^B e^{j2\pi f\tau} df \\ &= \frac{N_0 \sin(2\pi B\tau)}{2\pi\tau}. \end{aligned}$$

With $\text{sinc}(x) = \sin(\pi x) / (\pi x)$ **and its continuous value at zero,**

$$R_{N_o}(\tau) = N_0 B \text{sinc}(2B\tau).$$

Zero-lag check

$$P_{N_o} = R_{N_o}(0) = N_0 B.$$

The first nonzero-lag zeros are $\tau = \pm 1 / (2B)$.

Band-pass filtering of triangular-spectrum noise

Question

Zero-mean noise has the two-sided PSD

$$S_{N_i}(f) = \begin{cases} 10^{-8} \left(1 - \frac{|f|}{10^8}\right), & |f| < 10^8 \text{ Hz}, \\ 0, & \text{otherwise.} \end{cases}$$

A unity-gain ideal band-pass filter has a positive-frequency passband of width 2 MHz centered at 50 MHz, and a mirrored negative-frequency passband.

Determine:

- The output PSD.
- The input and output noise powers.

Solution

Identify the passbands

The positive-frequency edges are 49 MHz and 51 MHz. Therefore,

$$S_{N_o}(f) = \begin{cases} 10^{-8} \left(1 - \frac{|f|}{10^8}\right), & 49 \times 10^6 \leq |f| \leq 51 \times 10^6, \\ 0, & \text{otherwise.} \end{cases}$$

Input power

$$P_{N_i} = 2 \times 10^{-8} \int_0^{10^8} \left(1 - \frac{f}{10^8}\right) df = 1.$$

Output power

$$\begin{aligned} P_{N_o} &= 2 \times 10^{-8} \int_{49 \times 10^6}^{51 \times 10^6} \left(1 - \frac{f}{10^8}\right) df \\ &= 2 \times 10^{-8} \left[f - \frac{f^2}{2 \times 10^8} \right]_{49 \times 10^6}^{51 \times 10^6} \\ &= 0.02. \end{aligned}$$

Thus $100P_{N_o}/P_{N_i} = 2\%$. If the PSD is in W/Hz, the powers are in watts.

Triangular-spectrum noise through a ramp-gain filter

Question

Zero-mean noise has the two-sided PSD

$$S_{N_i}(f) = \begin{cases} 10^{-8} \left(1 - \frac{|f|}{10^8}\right), & |f| < 10^8 \text{ Hz}, \\ 0, & \text{otherwise.} \end{cases}$$

The filter has

$$|H(f)|^2 = \begin{cases} |f| - 49 \times 10^6, & 49 \times 10^6 \leq |f| \leq 51 \times 10^6, \\ 0, & \text{otherwise.} \end{cases}$$

Evaluate f numerically in hertz and use the specified gain without normalization.

Determine:

- The output PSD.
- The output noise power.
- The output-to-input power ratio.

Solution

Output spectrum

$$S_{N_o}(f) = \begin{cases} 10^{-8} \left(1 - \frac{|f|}{10^8}\right) (|f| - 49 \times 10^6), & 49 \times 10^6 \leq |f| \leq 51 \times 10^6, \\ 0, & \text{otherwise.} \end{cases}$$

Integrate both passbands

Set $a = 49 \times 10^6$, $L = 2 \times 10^6$ and $u = f - a$. Then

$$\begin{aligned} P_{N_o} &= 2 \times 10^{-8} \int_0^L \left(0.51 - \frac{u}{10^8}\right) u \, du \\ &= 2 \times 10^{-8} \left[\frac{0.51L^2}{2} - \frac{L^3}{3 \times 10^8} \right] \\ &= \frac{59600}{3} \approx 1.98667 \times 10^4. \end{aligned}$$

Power ratio

The input triangular spectrum has area $P_{N_i} = 1$. Hence

$$\frac{P_{N_o}}{P_{N_i}} = \frac{59600}{3}, \quad 100 \frac{P_{N_o}}{P_{N_i}} \approx 1.98667 \times 10^6\%.$$

The specified maximum squared gain is 2×10^6 . Powers are in watts when the input PSD is in W/Hz.

Autocorrelation of ideal band-pass filtered white noise

Question

White noise with two-sided PSD $N_0/2$ passes through an ideal band-pass filter. Its magnitude is K on each interval $|f - f_c| \leq B$ and $|f + f_c| \leq B$, and zero elsewhere, with $f_c > B > 0$.

Determine the output-noise autocorrelation.

Solution

Output spectrum

Each passband has width $2B$:

$$S_{N_o}(f) = \frac{K^2 N_0}{2} \left[\text{rect} \left(\frac{f - f_c}{2B} \right) + \text{rect} \left(\frac{f + f_c}{2B} \right) \right].$$

Inverse Fourier transform

Using the rectangular-spectrum transform and frequency translation,

$$\begin{aligned}
 R_{N_o}(\tau) &= K^2 N_0 B \operatorname{sinc}(2B\tau) \left(e^{j2\pi f_c \tau} + e^{-j2\pi f_c \tau} \right) \\
 &= 2K^2 N_0 B \operatorname{sinc}(2B\tau) \cos(2\pi f_c \tau).
 \end{aligned}$$

Therefore,

$$R_{N_o}(\tau) = 2K^2 N_0 B \operatorname{sinc}(2B\tau) \cos(2\pi f_c \tau).$$

Here $\operatorname{sinc}(x) = \sin(\pi x) / (\pi x)$, with $\operatorname{sinc}(0) = 1$.

Power check

$$P_{N_o} = R_{N_o}(0) = 2K^2 N_0 B.$$

First-order filtering with a 10 rad/s cutoff

Question

White noise has $R_X(\tau) = (N_0/2) \delta(\tau)$ and drives

$$H(\omega) = \frac{1}{1 + j\omega/10}, \quad \omega = 2\pi f.$$

Determine:

- The impulse response.
- The output-noise autocorrelation.

Solution

Impulse response

$$H(\omega) = \frac{10}{10 + j\omega} \Rightarrow h(t) = 10e^{-10t}u(t).$$

Output spectrum

$$S_Y(f) = \frac{N_0}{2} |H(f)|^2 = \frac{N_0}{2} \frac{100}{100 + (2\pi f)^2}.$$

Output autocorrelation

Using $e^{-a|\tau|} \leftrightarrow 2a / [a^2 + (2\pi f)^2]$ with $a = 10$,

$$R_Y(\tau) = \frac{5N_0}{2} e^{-10|\tau|}.$$

The output power is $P_Y = 5N_0/2$, and the cutoff frequency is $10/(2\pi)$ Hz.

Finite-band powers after first-order noise filtering

Question

A zero-mean white Gaussian input has $S_X(f) = \eta/2$ and drives a filter with $h(t) = e^{-t}u(t)$.

Determine:

- The output PSD, autocorrelation and mean-square value.
- The input and output powers in the band $|f| \leq B$, with positive low-pass edge $B = 10^3$ Hz.
- The percentage of total output power within that band and the output-to-input power percentage within the same band.

Solution

Frequency response and output PSD

$$\begin{aligned} H(f) &= \int_0^\infty e^{-t} e^{-j2\pi ft} dt = \frac{1}{1+j2\pi f}, \\ S_Y(f) &= \frac{\eta}{2} |H(f)|^2. \end{aligned}$$

Therefore,

$$S_Y(f) = \frac{\eta}{2 [1 + (2\pi f)^2]}.$$

Autocorrelation and mean square

The exponential Fourier pair gives

$$R_Y(\tau) = \frac{\eta}{4} e^{-|\tau|}, \quad E\{Y^2(t)\} = P_{Y,\text{total}} = \frac{\eta}{4}.$$

Powers in the measurement band

$$\begin{aligned} P_{X,B} &= \int_{-B}^B \frac{\eta}{2} df = \eta B = 1000\eta, \\ P_{Y,B} &= \frac{\eta}{2} \int_{-B}^B \frac{df}{1 + (2\pi f)^2} = \frac{\eta}{2\pi} \tan^{-1}(2\pi B). \end{aligned}$$

Power percentages

At $B = 10^3$ Hz,

$$100 \frac{P_{Y,B}}{P_{Y,\text{total}}} = 100 \frac{2}{\pi} \tan^{-1} (2\pi B) \approx 99.9899\%,$$

$$100 \frac{P_{Y,B}}{P_{X,B}} = 100 \frac{\tan^{-1} (2\pi B)}{2\pi B} \approx 0.0249975\%.$$

The first ratio measures how much of the total filtered noise lies in the measurement band; the second compares output and input powers over that same band.

Narrowband Noise and Its Quadrature Components

A time-domain approximation

A real zero-mean stationary Gaussian narrowband process has mirrored spectral lobes near $\pm f_c$, with bandwidth small compared with f_c . Consider first a rectangular spectral pair of height S_0 , each lobe having full width Δf and center $\pm f_c$. Then

$$R_Y(\tau) = 2S_0\Delta f \operatorname{sinc}(\Delta f\tau) \cos(2\pi f_c\tau).$$

For $|\tau| \ll 1/\Delta f$,

$$R_Y(\tau) \approx 2S_0\Delta f \cos(2\pi f_c\tau).$$

This correlation is reproduced by

$$\tilde{Y}(t) = A \cos(2\pi f_c t) + B \sin(2\pi f_c t),$$

where A, B are independent zero-mean Gaussian variables with

$$E\{A^2\} = E\{B^2\} = 2S_0\Delta f.$$

A general narrowband spectrum can be divided into thin spectral pairs. With pair centers $\pm f_m$ and width Δf ,

$$Y(t) \approx \sum_m [A_m \cos(2\pi f_m t) + B_m \sin(2\pi f_m t)],$$

$$E\{A_m^2\} = E\{B_m^2\} = 2S_Y(f_m) \Delta f.$$

Coefficients from different pairs are uncorrelated and jointly Gaussian. The resulting mean square is the spectral-area approximation

$$E\{Y^2(t)\} \approx \sum_m 2S_Y(f_m) \Delta f.$$

In-phase and quadrature representation

The same noise can be written using slowly varying baseband components:

$$Y(t) = Y_c(t) \cos(2\pi f_c t) - Y_s(t) \sin(2\pi f_c t).$$

Y_c is the in-phase component and Y_s the quadrature component. For zero-mean stationary Gaussian bandpass noise, they are zero-mean, low-pass, jointly stationary Gaussian processes.

Translate the positive lobe left by f_c and the negative lobe right by f_c , then add:

$$S_{Y_c}(f) = S_{Y_s}(f) = S_Y(f + f_c) + S_Y(f - f_c),$$

within the retained baseband support. With

$$R_{Y_c Y_s}(\tau) = E\{Y_c(t + \tau) Y_s(t)\},$$

the cross spectra are

$$S_{Y_c Y_s}(f) = j[S_Y(f + f_c) - S_Y(f - f_c)],$$

$$S_{Y_s Y_c}(f) = S_{Y_c Y_s}^*(f).$$

Their imaginary, odd shape gives $R_{Y_c Y_s}(0) = 0$. The component powers satisfy

$$P_{Y_c} = P_{Y_s} = P_Y = \int_{-\infty}^{\infty} S_Y(f) df.$$

For real components,

$$R_Y(\tau) = R_{Y_c}(\tau) \cos(2\pi f_c \tau) + R_{Y_c Y_s}(\tau) \sin(2\pi f_c \tau).$$

If the lobes are symmetric about the chosen carrier, the cross spectrum vanishes at every frequency. The components are then independent Gaussian processes, and

$R_Y = R_{Y_c} \cos(2\pi f_c \tau)$. Without that symmetry, the same-time pair remains independent, but different-time component values can be correlated.

Example: Symmetric bandpass white noise

Question

White Gaussian noise of PSD $N_0/2$ passes through a unity-gain filter with positive passband $[f_c - B, f_c + B]$ and its negative-frequency mirror, where $0 < B \ll f_c$. Find the noise and component spectra, correlations, and powers.

Solution

Each lobe has full width $2B$, so

$$P_Y = 2 \left(2B \frac{N_0}{2} \right) = 2N_0 B.$$

The translated lobes coincide at baseband:

$$S_{Y_c}(f) = S_{Y_s}(f) = \begin{cases} N_0, & |f| \leq B, \\ 0, & |f| > B. \end{cases}$$

Thus

$$R_{Y_c}(\tau) = R_{Y_s}(\tau) = 2N_0B \operatorname{sinc}(2B\tau),$$

$$R_Y(\tau) = 2N_0B \operatorname{sinc}(2B\tau) \cos(2\pi f_c \tau),$$

$$P_{Y_c} = P_{Y_s} = P_Y = 2N_0B.$$

Each component has the same power as the bandpass process. Sinusoidal reconstruction contributes a factor one-half to each component's power.

Example: An asymmetric triangular spectrum

Question

A zero-mean Gaussian process has a positive triangular lobe with vertices $(4, 0)$, $(5, 1)$, $(7, 0)$, with frequency in hertz, and a reflected negative lobe. Find the component spectra at reference carrier $f_c = 5$ Hz and the same-time joint density.

Solution

The total spectral area is

$$P_Y = 2 \left[\frac{1}{2} (7 - 4) \right] = 3.$$

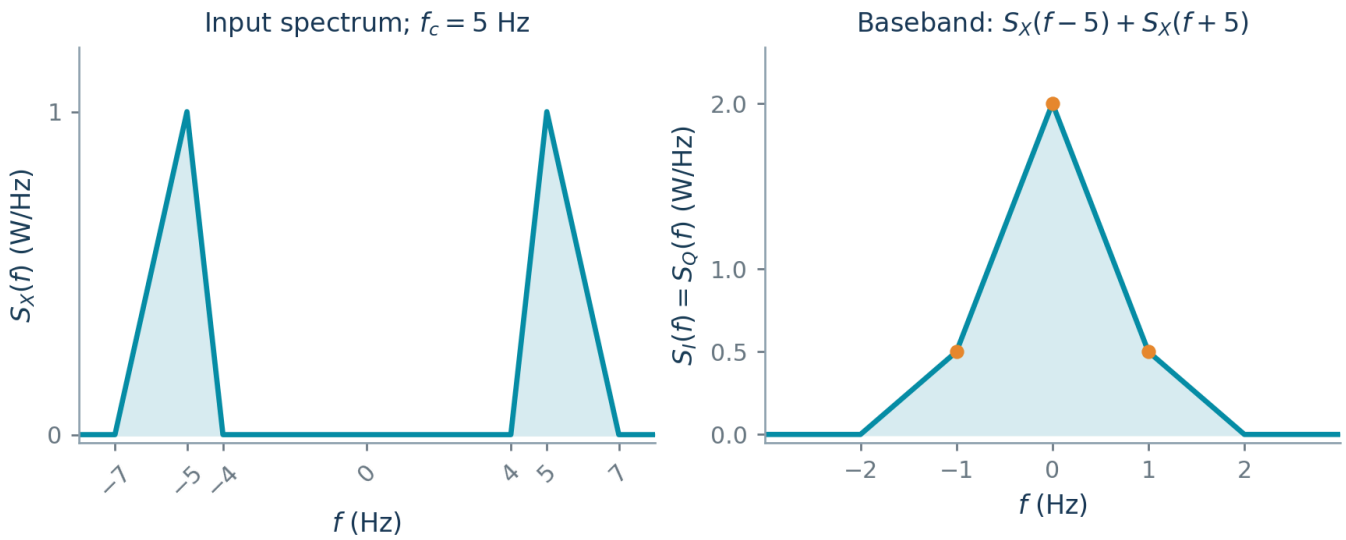
Adding the two translated shapes gives

$$S_{Y_c}(f) = S_{Y_s}(f) = \begin{cases} 2 - \frac{3}{2}|f|, & |f| \leq 1, \\ 1 - \frac{1}{2}|f|, & 1 < |f| \leq 2, \\ 0, & |f| > 2. \end{cases}$$

The peak is two, and each component spectrum has area three. At a common time, the Gaussian pair has zero covariance and variances three:

$$f_{Y_c, Y_s}(u, v) = \frac{1}{6\pi} \exp \left[-\frac{u^2 + v^2}{6} \right].$$

The spectrum's asymmetry affects unequal-time cross-correlation, while the same-time pair is independent.



A bandpass triangle and its shifted-and-added component spectrum.

Envelope and phase

Define

$$R(t) = \sqrt{Y_c^2(t) + Y_s^2(t)}, \quad \Theta(t) = \text{atan2}(Y_s(t), Y_c(t)).$$

Then

$$Y(t) = R(t) \cos[2\pi f_c t + \Theta(t)].$$

At a fixed time, write $\sigma^2 = P_Y > 0$. The independent Gaussian components have joint density

$$f_{Y_c, Y_s}(u, v) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{u^2 + v^2}{2\sigma^2}\right].$$

Substitute $u = r \cos \theta$, $v = r \sin \theta$ with Jacobian magnitude r :

$$f_{R, \Theta}(r, \theta) = \frac{r}{2\pi\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0, \quad -\pi < \theta \leq \pi.$$

It factors into the Rayleigh envelope density and uniform phase density:

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \quad r \geq 0, \quad f_\Theta(\theta) = \frac{1}{2\pi}.$$

The envelope and phase are independent at a fixed time. Their useful envelope moments are

$$E\{R\} = \sigma\sqrt{\frac{\pi}{2}}, \quad E\{R^2\} = 2\sigma^2, \quad \text{Var}(R) = \left(2 - \frac{\pi}{2}\right)\sigma^2.$$

Worked Problems

Envelope and phase of a carrier plus narrowband noise

Question

A carrier $x(t) = A \cos(2\pi f_c t)$ is added to narrowband noise

$$n(t) = n_c(t) \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t).$$

Express $y(t) = x(t) + n(t)$ in the form

$$y(t) = \rho(t) \cos[2\pi f_c t + \theta(t)].$$

Solution

Combine the quadrature components

$$y(t) = [A + n_c(t)] \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t).$$

Match the envelope-phase coefficients

Expanding $\rho(t) \cos[2\pi f_c t + \theta(t)]$ gives

$$\begin{aligned} \rho(t) \cos \theta(t) &= A + n_c(t), \\ \rho(t) \sin \theta(t) &= n_s(t). \end{aligned}$$

Hence

$$\rho(t) = \sqrt{[A + n_c(t)]^2 + n_s^2(t)},$$

$$\theta(t) = \text{atan2}(n_s(t), A + n_c(t)).$$

The two-argument arctangent selects the correct quadrant. When $\rho(t) = 0$, phase may be assigned arbitrarily.

I/Q spectra of asymmetric triangular-band noise

Question

The PSD $S_n(f)$ has a positive-frequency triangular lobe supported on $4 \leq f \leq 7$ Hz, with height 1 at $f = 5$ Hz. It rises linearly from zero at 4 Hz and falls linearly to zero at 7 Hz. The negative-frequency lobe is its mirror.

Use $f_c = 5$ Hz and

$$n(t) = n_c(t) \cos(2\pi f_c t) - n_s(t) \sin(2\pi f_c t).$$

Determine:

- The in-phase and quadrature PSDs.
- Both cross-spectral densities, using $R_{XY}(\tau) = E\{X(t+\tau)Y(t)\}$.

Solution

Express the band-pass spectrum

$$T(f) = \begin{cases} f-4, & 4 \leq f \leq 5, \\ \frac{7-f}{2}, & 5 \leq f \leq 7, \\ 0, & \text{otherwise,} \end{cases} \quad S_n(f) = T(f) + T(-f).$$

Translate and add the lobes

$$S_{n_c}(f) = S_{n_s}(f) = S_n(f+5) + S_n(f-5).$$

For $r = |f|$, with f in hertz,

$$S_{n_c}(f) = S_{n_s}(f) = \begin{cases} 2 - \frac{3}{2}r, & 0 \leq r \leq 1, \\ 1 - \frac{1}{2}r, & 1 < r \leq 2, \\ 0, & r > 2. \end{cases}$$

Cross spectra

$$\begin{aligned} S_{n_c n_s}(f) &= j[S_n(f+5) - S_n(f-5)], \\ S_{n_s n_c}(f) &= S_{n_c n_s}^*(f). \end{aligned}$$

Define

$$q(r) = \begin{cases} \frac{r}{2}, & 0 \leq r \leq 1, \\ 1 - \frac{r}{2}, & 1 < r \leq 2, \\ 0, & r > 2. \end{cases}$$

The explicit answers are

$$S_{n_c n_s}(f) = j \operatorname{sgn}(f) q(|f|), \quad S_{n_s n_c}(f) = -j \operatorname{sgn}(f) q(|f|).$$

Power and covariance checks

$$\begin{aligned} P_{n_c} = P_{n_s} &= 2 \left[\int_0^1 \left(2 - \frac{3}{2}r\right) dr + \int_1^2 \left(1 - \frac{1}{2}r\right) dr \right] \\ &= 3. \end{aligned}$$

The imaginary odd cross spectrum integrates to zero, so $C_{n_c n_s}(0) = 0$. It is not identically zero, so unequal-time cross-correlation can remain.

I/Q spectra with three carrier choices

Question

Zero-mean white noise has two-sided PSD $N_0/2$ and passes through a filter with

$$|H(f)|^2 = \begin{cases} \frac{1}{2}, & 10 \leq |f|/10^3 < 12, \\ 1, & 12 \leq |f|/10^3 \leq 14, \\ 0, & \text{otherwise,} \end{cases}$$

where f is in hertz.

Determine:

- The output PSD and total noise power.
- The I/Q PSDs and both cross spectra for $f_c = 10, 12, 14$ kHz.

Use $n = n_c \cos(2\pi f_c t) - n_s \sin(2\pi f_c t)$ and $R_{XY}(\tau) = E\{X(t+\tau)Y(t)\}$.

Solution

Output PSD and power

$$S_n(f) = \begin{cases} \frac{N_0}{4}, & 10 \leq |f|/10^3 < 12, \\ \frac{N_0}{2}, & 12 \leq |f|/10^3 \leq 14, \\ 0, & \text{otherwise.} \end{cases}$$

Both lobes contribute, with each subband width equal to 2000 Hz:

$$P_n = 2 \left[\frac{N_0}{4} (2000) + \frac{N_0}{2} (2000) \right] = 3000N_0.$$

Shifted-spectrum formulas

Let $S_I(f) = S_{n_c}(f) = S_{n_s}(f)$ and $r = |f|/10^3$. Then

$$\begin{aligned} S_I(f) &= S_n(f + f_c) + S_n(f - f_c), \\ S_{n_c n_s}(f) &= j [S_n(f + f_c) - S_n(f - f_c)], \\ S_{n_s n_c}(f) &= S_{n_c n_s}^*(f). \end{aligned}$$

Carrier $f_c = 10$ kHz

$$S_I(f) = \begin{cases} \frac{N_0}{4}, & 0 < r < 2, \\ \frac{N_0}{2}, & 2 < r < 4, \\ 0, & r > 4, \end{cases}$$

$$S_{n_c n_s}(f) = j \operatorname{sgn}(f) S_I(f).$$

Carrier $f_c = 12$ kHz

$$S_I(f) = \begin{cases} \frac{3N_0}{4}, & r < 2, \\ 0, & r > 2, \end{cases}$$

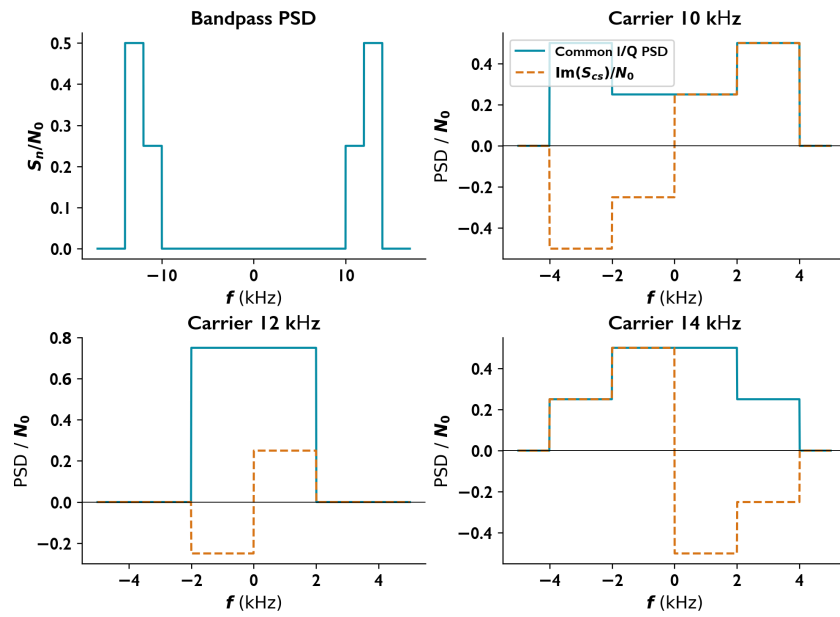
$$S_{n_c n_s}(f) = \begin{cases} j \operatorname{sgn}(f) \frac{N_0}{4}, & 0 < r < 2, \\ 0, & r > 2. \end{cases}$$

Carrier $f_c = 14$ kHz

$$S_I(f) = \begin{cases} \frac{N_0}{2}, & 0 < r < 2, \\ \frac{N_0}{4}, & 2 < r < 4, \\ 0, & r > 4, \end{cases}$$

$$S_{n_c n_s}(f) = -j \operatorname{sgn}(f) S_I(f).$$

For every carrier, $S_{n_s n_c} = S_{n_c n_s}^*$ and each component power is $3000N_0$. Values at isolated spectral breakpoints do not affect these powers.



Frequency axes are in kHz. The dashed orange curve is the imaginary cross spectrum under the stated first-variable lag convention.

Modulation and noise

Sinusoidal power and AM sidebands For a unit resistance, a sinusoid of amplitude A has power $A^2/2$.

$$\begin{aligned}s_{\text{AM}}(t) &= A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t), \\s_{\text{AM}}(t) &= A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2} \cos[2\pi(f_c + f_m)t] + \frac{A_c \mu}{2} \cos[2\pi(f_c - f_m)t], \\P_{\text{AM}} &= \frac{A_c^2}{2} \left(1 + \frac{\mu^2}{2}\right).\end{aligned}$$

After filtering, multiply each sinusoidal amplitude by its filter gain before calculating power. For DSB-SC,

$$s(t) = A_c m(t) \cos(2\pi f_c t), \quad P_s = \frac{A_c^2 P_m}{2}.$$

Filtered and coherently translated noise Let $N_0/2$ be the two-sided white-noise PSD.

$$S_n(f) = \frac{N_0}{2} |H(f)|^2, \quad P_n = \int_{-\infty}^{\infty} S_n(f) df.$$

For multiplication by $A_L \cos(2\pi f_c t)$ followed by a lowpass filter, let $\kappa = A_L/2$. For separated passbands,

$$S_{n,o}(f) = \kappa^2 [S_n(f - f_c) + S_n(f + f_c)]$$

within the output passband. Ideal DSB-SC reception gives $m_o(t) = \kappa A_c m(t)$. For AM, remove carrier-generated DC when finding message SNR.

$$\text{SNR} = \frac{P_s}{P_n}, \quad \text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}).$$

FM bandwidth, power, and threshold Assume a unit-peak message, ideal above-threshold reception, and no pre-emphasis or de-emphasis.

$$\beta = \frac{\Delta f}{W}, \quad B_T \simeq 2(\beta + 1)W,$$

$$\text{SNR}_o = \frac{3\beta^2 P_m P_r}{N_0 W}, \quad P_r = P_T 10^{-L/10}.$$

Here W is the message bandwidth, P_m its mean-square value, and L the channel power loss in decibels. If the threshold ratio is ten,

$$P_{n,\text{BP}} = N_0 B_T, \quad \frac{P_r}{N_0 B_T} \geq 10.$$

Worked Problems

AM signal and noise through a nonuniform bandpass filter

Question

A carrier with $A_c = 10$ mV and $f_c = 90$ kHz is 50% amplitude modulated by a 750 Hz sinusoid. The two-sided noise PSD is $\eta/2 = 10^{-3}$ W/Hz.

The filter magnitude is even; on the positive-frequency lobe,

$$|H(f)| = \begin{cases} 1, & 89.5 \leq f \leq 90.5 \text{ kHz}, \\ 0.5, & 89 \leq f < 89.5 \text{ kHz or } 90.5 < f \leq 91 \text{ kHz}, \\ 0, & \text{otherwise.} \end{cases}$$

The output is multiplied by $2 \cos(2\pi f_c t)$ and passed through an ideal LPF with bandwidth 1 kHz.

Find the demodulator input SNR and the recovered-message output SNR.

Solution

Assume coherent phase at the carrier and sidebands, and a unit resistance.

Input signal and noise With $\mu = 0.5$, each sideband initially has amplitude $A_c\mu/2$. The carrier gain is one; the sideband gains are $1/2$.

$$\begin{aligned} P_{s,i} &= \frac{A_c^2}{2} + 2 \frac{1}{2} \left(\frac{A_c\mu}{4} \right)^2 = 5.15625 \times 10^{-5} \text{ W}, \\ \int |H(f)|^2 df &= 2 \left[1000 + 2 (500) (0.5)^2 \right] = 2500 \text{ Hz}, \\ P_{n,i} &= 10^{-3} (2500) = 2.5 \text{ W}. \end{aligned}$$

$$\text{SNR}_i = 2.0625 \times 10^{-5} \approx -46.86 \text{ dB}$$

Recovered message

$$s_o(t) = A_c + \frac{A_c\mu}{2} \cos(2\pi f_m t), \quad P_{m,o} = \frac{1}{2} \left(\frac{A_c\mu}{2} \right)^2 = 3.125 \times 10^{-6} \text{ W}.$$

Exclude the carrier-generated DC term from message power.

Output noise

$$S_{n,o}(f) = \frac{\eta}{2} \left[|H(f - f_c)|^2 + |H(f + f_c)|^2 \right].$$

$$S_{n,o}(f) = \begin{cases} 2 \times 10^{-3}, & |f| \leq 500, \\ 0.5 \times 10^{-3}, & 500 < |f| \leq 1000, \\ 0, & \text{otherwise.} \end{cases}$$

$$P_{n,o} = (2 \times 10^{-3}) (1000) + (0.5 \times 10^{-3}) (1000) = 2.5 \text{ W}.$$

$$\text{SNR}_o = 1.25 \times 10^{-6} \approx -59.03 \text{ dB}$$

If DC is included, $(A_c^2 + P_{m,o}) / P_{n,o} = 4.125 \times 10^{-5}$.

FM power, Carson bandwidth, and threshold limitation

Question

An audio signal has $W = 15 \text{ kHz}$, average power $P_m = 1/6$, and peak magnitude one. FM transmission experiences 60 dB power loss and two-sided AWGN PSD $\eta/2 = 10^{-12} \text{ W/Hz}$. The required output SNR is at least 40 dB.

- Find the bandwidth and transmitter power for deviation ratio $\beta = 5$.
- Find the maximum deviation ratio and bandwidth when the carrier power must be at least ten times the channel noise power.

Solution

Assume ideal above-threshold FM without pre-emphasis or de-emphasis. Let $N_0 = \eta = 2 \times 10^{-12}$.

Bandwidth and power

$$B_T \simeq 2(\beta + 1)W = 180 \text{ kHz}$$

$$\text{SNR}_o = \frac{3\beta^2 P_m P_r}{N_0 W} = 10^4,$$

$$P_r = \frac{10^4 (2 \times 10^{-12}) (15000)}{3(25)(1/6)} = 2.4 \times 10^{-5} \text{ W},$$

$$P_T = 10^6 P_r.$$

$$P_T = 24 \text{ W}$$

Here $N_0 B_T = 3.6 \times 10^{-7} \text{ W}$ and $P_r / (N_0 B_T) = 66.67 > 10$.

Threshold at fixed transmitter power Keep $P_T = 24 \text{ W}$.

$$\frac{P_r}{N_0 B_T} \geq 10,$$

$$B_T \leq \frac{P_r}{10 N_0} = 1.2 \times 10^6 \text{ Hz},$$

$$\beta \leq \frac{1.2 \times 10^6}{2(15000)} - 1.$$

$$\beta_{\max} = 39, \quad B_{T, \max} = 1.2 \text{ MHz}$$

These are boundary values; strictly remaining above threshold requires a smaller bandwidth.

Power retuned for each deviation ratio If power is instead minimized anew at each β while maintaining 40 dB output SNR,

$$P_r = \frac{6 \times 10^{-4}}{\beta^2}, \quad P_r \geq 6 \times 10^{-7} (\beta + 1).$$

$$\beta^2 (\beta + 1) \leq 1000.$$

$$\beta_{\max} = 9.67753, \quad B_T = 320.326 \text{ kHz}$$

Minimum FM transmitter power under a channel bandwidth limit

Question

A normalized message has bandwidth $W = 5 \text{ kHz}$ and power $P_m = 0.5$. The channel has bandwidth 60 kHz, 40 dB power attenuation, and two-sided noise PSD 10^{-9} W/Hz .

For FM without pre-emphasis or de-emphasis, find the minimum transmitter power and deviation ratio needed for at least 40 dB output SNR.

Solution

Assume unit peak message magnitude and ideal above-threshold FM.

Maximum deviation

$$2 (\beta + 1) (5000) \leq 60000 \Rightarrow \beta \leq 5.$$

Since required power is proportional to $1/\beta^2$,

$$\beta = 5, \quad \Delta f = 25 \text{ kHz}$$

Minimum power The two-sided density gives $N_0 = 2 \times 10^{-9}$.

$$\begin{aligned} \text{SNR}_o &= \frac{3\beta^2 P_m P_r}{N_0 W} = 10^4, \\ P_{r, \min} &= \frac{10^4 (2 \times 10^{-9}) (5000)}{3 (25) (0.5)} = 2.66667 \times 10^{-3} \text{ W}, \\ P_{T, \min} &= 10^4 P_{r, \min}. \end{aligned}$$

$$P_{T, \min} = 26.667 \text{ W}$$

The predetection ratio is

$$\frac{P_r}{N_0 B_T} = \frac{2.66667 \times 10^{-3}}{(2 \times 10^{-9}) (60000)} = 22.22,$$

above a ten-to-one threshold criterion.

DSB-SC filtering and coherent-detection SNR

Question

The transmitted signal is

$$s(t) = 10 \left[3 \cos(2\pi 10^2 t) + 2 \cos(2\pi 10^4 t) \right] \cos(\pi 10^8 t).$$

White noise has two-sided PSD $N_0/2 = 2 \times 10^{-6}$ W/Hz. A unity-gain ideal band-pass filter is centered at the carrier and has full positive-frequency passband width 10 kHz, with the mirrored negative-frequency passband.

Determine:

- The modulation type.
- The SNR at the band-pass filter output.
- The in-phase noise PSD and average power.
- The SNR after coherent detection with multiplier $2 \cos(2\pi f_c t)$ and low-pass edge 5 kHz.

Use a unit-resistance power normalization.

Solution

Carrier frequency and passed signal

The modulation is double-sideband suppressed carrier, with

$$f_c = \frac{10^8}{2} = 5 \times 10^7 \text{ Hz}.$$

The positive passband edges are $f_c \pm B$, with $B = 5000$ Hz. The 100 Hz tone passes; the 10 kHz tone is rejected. Therefore,

$$s_o(t) = 30 \cos(2\pi 100t) \cos(2\pi f_c t).$$

Its mean-square power is

$$P_s = 30^2 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 225 \text{ W}.$$

Band-pass noise and SNR

The combined width of both passbands is $4B$:

$$P_n = \frac{N_0}{2} (4B) = 0.04 \text{ W}.$$

Hence

$$\text{SNR}_{\text{BPF}} = \frac{225}{0.04} = 5625 \approx 37.50 \text{ dB}.$$

In-phase noise

For $n = n_c \cos (2\pi f_c t) - n_s \sin (2\pi f_c t)$,

$$S_{n_c}(f) = \begin{cases} 4 \times 10^{-6} \text{ W/Hz}, & |f| \leq 5000 \text{ Hz}, \\ 0, & \text{otherwise.} \end{cases}$$

$$P_{n_c} = \int_{-5000}^{5000} 4 \times 10^{-6} df = 0.04 \text{ W}.$$

Coherent detector output

Mixing by $2 \cos (2\pi f_c t)$ and retaining $|f| \leq 5000 \text{ Hz}$ gives

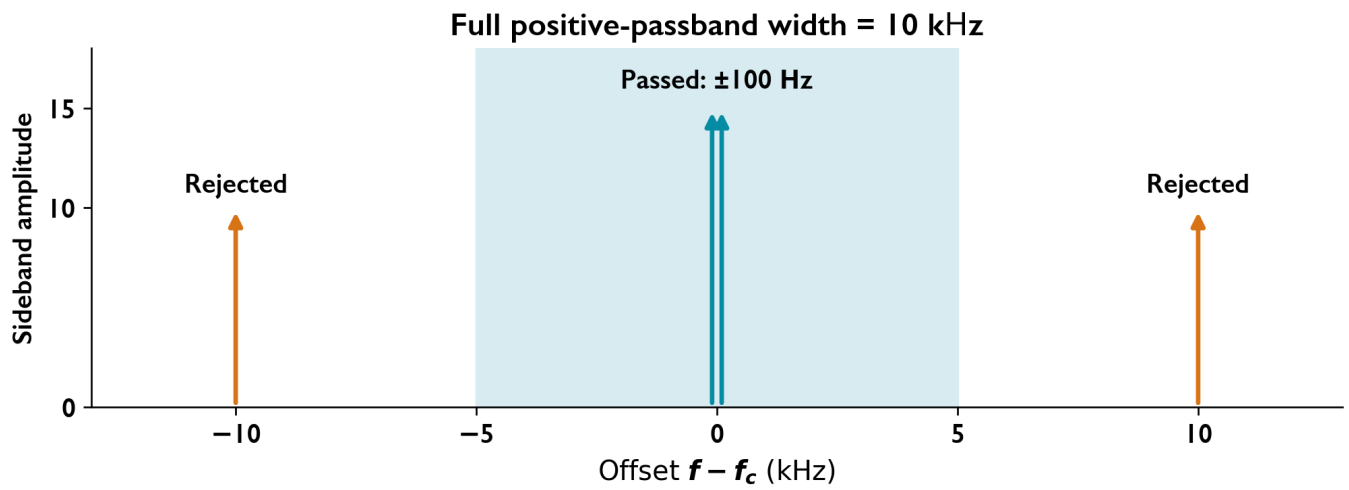
$$y_d(t) = 30 \cos (2\pi 100t) + n_c(t).$$

Thus

$$P_{s,d} = \frac{30^2}{2} = 450 \text{ W}, \quad P_{n,d} = 0.04 \text{ W},$$

$$\text{SNR}_d = \frac{450}{0.04} = 11250 \approx 40.51 \text{ dB}.$$

For a narrower detector edge B_d , with $100 \leq B_d \leq 5000 \text{ Hz}$, the signal power remains 450 W and the noise power becomes $2N_0B_d$.



Positive-frequency sidebands relative to the carrier. The reflected negative-frequency band also contributes noise power.

Sampling, quantization, and prediction

Samples and correlation

$$X[n] = X(nT_s), \quad T_s = \frac{1}{f_s}, \quad r_X[k] = R_X(kT_s).$$

A PSD specifies sample statistics; it does not specify individual random sample amplitudes.

Exact quantization powers Let bin $\mathcal{B}_i$ have reconstruction level q_i .

$$\begin{aligned} p_i &= \int_{\mathcal{B}_i} f_X(x) dx, \\ P_X &= \int x^2 f_X(x) dx, \\ P_q &= \sum_i \int_{\mathcal{B}_i} (x - q_i)^2 f_X(x) dx, \\ P_{\text{out}} &= \sum_i q_i^2 p_i, \quad \text{SQNR} = \frac{P_X}{P_q}. \end{aligned}$$

For a uniform input over the complete operating range and uniform bins without overload,

$$\Delta = \frac{2A}{Q}, \quad P_q = \frac{\Delta^2}{12}.$$

For a nonuniform density, use the bin integrals rather than assuming a uniform error. A valid density must be nonnegative and integrate to one.

Minimum mean-square prediction For a real zero-mean WSS sequence,

$$\widehat{X}[n] = aX[n-1], \quad D[n] = X[n] - \widehat{X}[n].$$

Orthogonality gives

$$a = \frac{r_X[1]}{r_X[0]}, \quad \sigma_D^2 = r_X[0] - \frac{r_X[1]^2}{r_X[0]}.$$

If $r_X[k] = \sigma_X^2 \rho^{|k|}$, a longer optimum predictor needs only its first tap:

$$a_1 = \rho, \quad a_2 = a_3 = 0, \quad \sigma_D^2 = \sigma_X^2 (1 - \rho^2).$$

For a Gaussian sequence, the innovations are independent Gaussian variables.

Differential quantization levels Under the specified Gaussian quantization model,

$$P_q = \frac{2\sigma_D^2}{Q^2}, \quad Q_{\min} = \left\lceil \sqrt{\frac{2\sigma_D^2}{P_{q, \max}}} \right\rceil.$$

For binary coding, $b = \lceil \log_2 Q_{\min} \rceil$ and the implemented number of levels is 2^b .

Worked Problems

Prediction of a sampled Gaussian process and differential

quantization

Question

A zero-mean Gaussian process has

$$S_X(f) = \frac{16}{1 + (16\pi f)^2}, \quad f_s = 1 \text{ Hz.}$$

A differential quantizer subtracts the predictor

$$\widehat{X}[n] = a_1 X[n-1] + a_2 X[n-2] + a_3 X[n-3].$$

For a Gaussian input of variance σ^2 and Q quantization levels, the quantization-noise power is $2\sigma^2/Q^2$.

- Find the number of levels required for noise power at most 4.32×10^{-4} .
- Determine the sampled-process values at $T_s, 2T_s, 3T_s, 4T_s$.

Solution

Correlation and optimum prediction Assume minimum mean-square prediction. With $T_s = 1$ s,

$$R_X(\tau) = e^{-|\tau|/8}, \quad \rho = e^{-1/8} = 0.882496903, \quad r_X[m] = \rho^{|m|}.$$

$$\begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} \rho \\ \rho^2 \\ \rho^3 \end{bmatrix} \Rightarrow a_1 = \rho, \quad a_2 = a_3 = 0.$$

$$X[n] = \rho X[n-1] + D[n], \quad \sigma_D^2 = 1 - e^{-1/4} = 0.221199217.$$

Quantization levels

$$\begin{aligned} \frac{2\sigma_D^2}{Q^2} &\leq 4.32 \times 10^{-4}, \\ Q &\geq \sqrt{\frac{2(1 - e^{-1/4})}{4.32 \times 10^{-4}}} = 32.0011007. \end{aligned}$$

$$Q_{\min} = 33$$

For $Q = 32$, $P_q = 4.3202972 \times 10^{-4}$, just above the bound. If Q must be a power of two, $Q = 64$, $b = 6$.

Sample statistics A PSD does not determine individual sample amplitudes. The available results are

$$E\{X[i]\} = 0, \quad \text{Var}(X[i]) = 1, \quad \text{Cov}(X[i], X[j]) = \rho^{|i-j|}, \quad i, j \in \{1, 2, 3, 4\}.$$

Given $X[0]$ and independent Gaussian innovations $D[k]$, actual samples follow from

$$X[n] = \rho^n X[0] + \sum_{k=1}^n \rho^{n-k} D[k].$$

The corresponding autocorrelation samples are

$$\begin{aligned} R_X(T_s) &= 0.882497, & R_X(2T_s) &= 0.778801, \\ R_X(3T_s) &= 0.687289, & R_X(4T_s) &= 0.606531. \end{aligned}$$

Exact quantization powers for uniform and parabolic inputs

Question

A midrise quantizer has thresholds $-2, 0, 2$, reconstruction levels $-3, -1, 1, 3$, and input range $[-4, 4]$:

$$q(x) = \begin{cases} -3, & -4 \leq x < -2, \\ -1, & -2 \leq x < 0, \\ 1, & 0 \leq x < 2, \\ 3, & 2 \leq x \leq 4. \end{cases}$$

- For a uniform input on $[-4, 4]$, find quantization-noise power and SQNR.
- For $f_X(x) = A(1 - x^2/c)$ on $|x| \leq 4$, zero otherwise, find the normalization, quantization-noise power, and quantized-signal power when the density is valid. The positive denominator is uncertain: test $c = 10$ and give the $c = 16$ results conditionally, together with the general valid model.

Solution

Uniform input

$$\Delta = 2, \quad P_X = \frac{4^2}{3} = \frac{16}{3}, \quad P_q = \frac{\Delta^2}{12} = \frac{1}{3}.$$

$$\text{SQNR} = \frac{16/3}{1/3} = 16 \approx 12.04 \text{ dB}$$

Validity and normalization For $c = 10$, $1 - x^2/10$ changes sign on the stated support. No nonzero A makes it a density. For positive c , validity requires $c \geq 16$.

$$1 = A \int_{-4}^4 \left(1 - \frac{x^2}{c}\right) dx = A \left(8 - \frac{128}{3c}\right), \quad A = \frac{3c}{8(3c - 16)}.$$

Exact noise power

$$P_q = 2A \left[\int_0^2 (x-1)^2 \left(1 - \frac{x^2}{c}\right) dx + \int_2^4 (x-3)^2 \left(1 - \frac{x^2}{c}\right) dx \right].$$

$$P_q = \frac{5c - 28}{5(3c - 16)}$$

Quantized-signal power

$$P_1 = P(0 < X < 2) = \frac{3c-4}{4(3c-16)},$$

$$P_2 = P(2 < X < 4) = \frac{3c-28}{4(3c-16)}.$$

$$E\{q(X)^2\} = 2(P_1 + 9P_2) = \frac{15c - 128}{3c - 16}$$

Conditional model $c = 16$

$$A = \frac{3}{16}, \quad P_q = \frac{13}{40} = 0.325, \quad E\{q(X)^2\} = \frac{7}{2} = 3.5$$

Here $P_1 = 11/32$, $P_2 = 5/32$, and $E\{X^2\} = 16/5 = 3.2$. **Exact quantization error can be correlated with X , so subtracting noise power from input power does not give output power.**

Matched filters and binary detection

Matched filter For a known pulse $s(t)$ in two-sided AWGN with PSD $N_0/2$,

$$E_s = \int |s(t)|^2 dt, \quad h(t) = K s^*(T-t).$$

For a real pulse,

$$y_s(t) = K R_s(t-T), \quad y_s(T) = K E_s.$$

$$\sigma_{n,o}^2 = \frac{N_0}{2} |K|^2 E_s, \quad \text{SNR}_{\max} = \frac{2E_s}{N_0}.$$

The gain $K \neq 0$ changes amplitude but not SNR.

Coherent binary error probabilities Let $T_b = 1/R_b$ and $E_b \simeq A^2 T_b / 2$ for a passband sinusoid of amplitude A .

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du.$$

For equiprobable, undistorted signaling,

$$P_{e,\text{BFSK}} = Q\left(\sqrt{\frac{E_b}{N_0}}\right), \quad P_{e,\text{BPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

The BFSK formula requires orthogonal signaling; $\Delta f T_b = 1$ is a suitable envelope-model spacing.

PSK bandwidth assumptions For binary PSK and positive-lobe channel width B ,

$$B = (1 + \alpha) R_b$$

with raised-cosine roll-off α . Rectangular pulses instead give the first-null approximation $B \simeq 2R_b$; their sidelobes extend beyond this interval.

MAP detection with Laplace noise For polar levels $\pm A$, priors p_+, p_- , and $f_N(n) = \frac{\lambda}{2} e^{-\lambda|n|}$, decide $+A$ if

$$\lambda (|v + A| - |v - A|) > \ln \frac{p_-}{p_+}.$$

If $\left| \ln \left(p_- / p_+ \right) \right| < 2\lambda A$,

$$\gamma = \frac{1}{2\lambda} \ln \frac{p_-}{p_+}, \quad P_e = \sqrt{p_+ p_-} e^{-\lambda A}.$$

The average signal power is A^2 and the noise variance is $2/\lambda^2$, so

$$\text{SNR} = \frac{(\lambda A)^2}{2}.$$

Worked Problems

Matched filter for a flat-topped, linearly falling pulse

Question

The pulse is

$$s(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 2 - t, & 1 \leq t < 2, \\ 0, & \text{otherwise.} \end{cases}$$

The two-sided AWGN PSD is $(2/3) \times 10^{-3} \text{ W/Hz}$.

- Find and sketch a matched-filter impulse response and its signal output.
- Find the time and magnitude of the maximum output.
- Determine the maximum output SNR.

Solution

Matched impulse response Choose unit gain and sampling time $T = 2$.

$$h(t) = s(2 - t) = \begin{cases} t, & 0 < t \leq 1, \\ 1, & 1 < t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Signal output Let $a = |t - 2|$. Then $y_s(t) = R_s(t - 2)$. For $0 \leq a \leq 1$,

$$R_s(a) = \int_a^1 du + \int_1^{1+a} (2 - u) du + \int_{1+a}^2 (2 - u)(2 - u + a) du.$$

For $1 < a \leq 2$, $R_s(a) = \int_a^2 (2 - u) du$. Hence

$$y_s(t) = \begin{cases} \frac{4}{3} - \frac{1}{2}a - \frac{1}{2}a^2 + \frac{1}{6}a^3, & 0 \leq a \leq 1, \\ \frac{1}{2}(2 - a)^2, & 1 < a \leq 2, \\ 0, & a > 2, \end{cases} \quad a = |t - 2|.$$

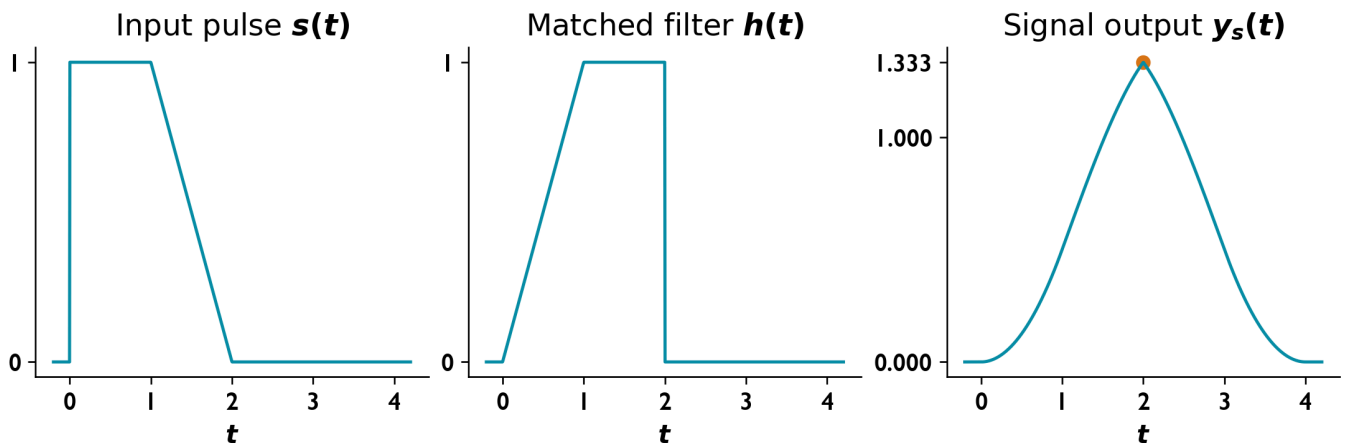
The output is symmetric about $t = 2$ and passes through

$$(t, y_s) = (0, 0), (1, \frac{1}{2}), (2, \frac{4}{3}), (3, \frac{1}{2}), (4, 0).$$

Peak and SNR

$$E_s = \int_0^1 dt + \int_1^2 (2 - t)^2 dt = \frac{4}{3}, \quad t_{\max} = 2, \quad |y_s(t_{\max})| = \frac{4}{3}.$$

$$\text{SNR}_{\max} = \frac{E_s}{N_0/2} = \frac{4/3}{(2/3) 10^{-3}} = 2000 \approx 33.01 \text{ dB}$$



Input pulse, causal matched-filter response, and signal output with peak $4/3$ at $t = 2$.

Coherent FSK and PSK over a finite bandpass channel

Question

A unity-gain channel passes $500 \leq |f| \leq 1500$ Hz and rejects all other frequencies. The amplitude-to-noise ratio is $A^2/\eta = 8000$.

- Find the coherent binary FSK error probability at 200 bit/s with tones at 1070 and 1270 Hz.
- Determine the fastest PSK rate and its coherent error probability.

Solution

Assume equiprobable bits, two-sided AWGN PSD $\eta/2 = N_0/2$, sinusoid amplitude A , and negligible distortion of the shaped signals.

Coherent FSK

$$\begin{aligned} T_b &= 1/200 = 0.005 \text{ s}, \\ (f_2 - f_1) T_b &= 200 (0.005) = 1, \\ \frac{E_b}{N_0} &\simeq \frac{A^2 T_b}{2\eta} = 8000 \frac{0.005}{2} = 20. \end{aligned}$$

The spacing satisfies envelope-model orthogonality.

$$P_{e,\text{FSK}} = Q(\sqrt{20}) = 3.8721 \times 10^{-6}$$

Binary PSK with ideal Nyquist shaping A maximum rate requires a pulse-shaping and modulation-order assumption. For binary PSK with zero roll-off, the positive-lobe width $B = 1000$ Hz equals R_b .

$$R_{b, \max} = 1000 \text{ bit/s}$$

$$\frac{E_b}{N_0} = \frac{A^2}{2\eta R_b} = 4,$$

$$P_{e, \text{BPSK}} = Q(\sqrt{8}) = 2.33887 \times 10^{-3}$$

For raised-cosine roll-off α , $R_{b, \max} = B / (1 + \alpha)$.

Rectangular-pulse bandwidth approximation Using the first-null approximation $B \simeq 2R_b$,

$$R_b \simeq 500 \text{ bit/s}, \quad E_b/N_0 = 8,$$

$$P_e \simeq Q(4) = 3.16712 \times 10^{-5}$$

This assumes undistorted coherent detection. A brick-wall channel truncates rectangular-pulse sidelobes, so the approximation is not an exact distortion analysis. Higher-order PSK requires a specified order.

MAP detection of polar data in Laplace noise

Question

At a binary receiver's sampling instant,

$$V_0(T_b) = s_0(T_b) + N_0(T_b), \quad f_N(n) = \frac{\lambda}{2} e^{-\lambda|n|}.$$

The polar levels are $+A$ for bit one and $-A$ for bit zero, with probabilities 0.25 and 0.75. Find the average SNR required for $P_e = 10^{-4}$ using the optimum threshold.

Solution

MAP threshold Decide $+A$ when

$$\frac{f_{V|+}(v)}{f_{V|-}(v)} = \exp\{\lambda(|v+A| - |v-A|)\} > 3.$$

Within $-A < v < A$, the log ratio is $2\lambda v$, giving

$$\gamma = \frac{\ln 3}{2\lambda}, \quad +A \text{ if } v > \gamma.$$

The interior condition is $\lambda A > (\ln 3)/2$.

Error probability

$$\begin{aligned} P_e &= \frac{1}{8} e^{-\lambda(A-\gamma)} + \frac{3}{8} e^{-\lambda(A+\gamma)} \\ &= \frac{\sqrt{3}}{4} e^{-\lambda A}. \end{aligned}$$

$$\lambda A = \ln \left(\frac{\sqrt{3}}{4 \times 10^{-4}} \right) = 8.37335216.$$

This satisfies the interior condition.

Average SNR

$$P_s = A^2, \quad \sigma_N^2 = \frac{2}{\lambda^2},$$

$$\text{SNR} = \frac{(\lambda A)^2}{2} = 35.0565 \approx 15.45 \text{ dB}.$$